Modern Portfolio Theory in the Crypto-Wilderness

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Modern portfolio theory (MPT) prescribes how to maximise the return of an asset portfolio for a given level of risk. The optimal trade-off between return and variance defines the efficient frontier. Whether actual portfolios approximate this prescription and whether proximity to the frontier translates into realised performance remain difficult to test at large scale in traditional markets due to their opaque nature and the inaccessibility of data, but become testable for cryptoasset portfolios. As we show, public blockchains make these questions measurable: every movement of the native currency and of every token is recorded, thus enabling complete portfolio reconstruction for every account at any point in time. We leverage this transparency to reconstruct cryptoasset portfolios of native ETH and ERC-20 tokens for over 116 M Ethereum accounts across the full chain history (2015–2025), measure their distance to the constrained efficient frontier, and quantify how deviations translate into realised performance. Here we show that market-wide movements, not allocation choice, dominate realised cryptoasset returns. On-chain wealth is highly concentrated and portfolios are under-diversified and largely static: 73.50% of accounts hold a single asset, and in a typical month 93.0% leave their holdings unchanged. Two-asset portfolios sit closest to the efficient frontier defined by their held assets, a proximity that reflects the narrowness of their opportunity set rather than deliberate optimisation. Passive market-capitalisation weighting outperforms every MPT strategy in median realised return, yet about half of the variation in returns (50.7–55.6%) comes from these market-wide movements, and frontier distance does not predict returns out of sample. Mean-variance optimisation therefore appears neither descriptive nor prescriptively useful for cryptoassets, even if MPT retains its value as a normative benchmark.

CCS Concepts: • **General and reference** → **Measurement**; • **Applied computing** → **Digital cash**; *Economics*; • **Information systems** → *Data analytics*; • **Mathematics of computing** → Mathematical optimization.

Additional Key Words and Phrases: Modern Portfolio Theory, Cryptoassets, Ethereum, DeFi, Blockchain

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1 Introduction

Modern portfolio theory (MPT), for which Markowitz received the Nobel Prize in Economics [38], prescribes that portfolios should lie on the mean-variance efficient frontier, that is, maximise expected return for a given level of risk. MPT's core assumptions are that investors are *perfectly*

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rational, actively optimising their allocation, and *risk-averse*, preferring lower variance for equal return. These assumptions are idealisations challenged in this paper.

Cryptoasset markets are a uniquely suitable empirical test bed for MPT. First, blockchains record every asset movement, of the native currency and of tokens alike, in a publicly auditable ledger, enabling the reconstruction of complete portfolio histories for every account, unlike traditional markets where holdings are observable only through selective brokerage disclosures. Second, cryptoassets are accessible permissionlessly, and on-chain portfolios reflect actual allocation decisions rather than constraints imposed by intermediaries or institutions. Third, the Ethereum ecosystem is particularly well-suited for this examination, spanning a decade of multiple market cycles and hosting the largest decentralised finance ecosystem. This setting enables us to examine both assumptions empirically at population scale.

In traditional markets, investors are systematically under-diversified and often underperform passive benchmarks [2, 9, 22], while on-chain studies reveal highly concentrated token distributions and centralised governance [41, 50]. Existing crypto-portfolio work applies MPT only to hypothetical asset baskets [7, 44, 45, 48] rather than to actual on-chain holdings, and each strand relies on sampled or selected data over limited windows. Yet no study has measured, at scale, how actual on-chain portfolios compare to MPT-optimal allocations, or whether deviating from the constrained efficient frontier has performance consequences.

We fill this gap with the first large-scale empirical study comparing *actual* on-chain cryptoasset portfolios to rebalanced *theoretical optimal* allocations under MPT. In our analysis, an account's *cryptoassets* comprise native ETH, the ecosystem's base currency, and ERC-20 tokens. We reconstruct these portfolios at scale and pursue three objectives: First (**Obj A**), we characterise the composition and diversification patterns of actual on-chain cryptoasset portfolios. Second (**Obj B**), we quantify the deviation of actual portfolios from the mean-variance efficient frontier under optimisation strategies and held-asset constraints. Third (**Obj C**), we examine how such deviations affect portfolio performance relative to the market. To summarise, we contribute the following:

(1) **Portfolio reconstruction method.** We provide a method for reconstructing complete on-chain portfolios, using transfer-like events from ERC-20 tokens and native ETH. We apply it over the full Ethereum history (2015–2025) to the 116 M accounts that have ever sent or received one of 4562 ERC-20 tokens, and reconstruct each account's full portfolio, including its native ETH. A ledger-based approach yields balances that match on-chain state exactly at randomly sampled accounts and blocks (Section 3). We contribute to open science by making the code (and the full dataset upon publication) publicly available[1].

(2) **Actual portfolio composition and diversification.** Aggregate on-chain wealth peaks at $1.91 T, but is highly concentrated: most accounts hold less than $100, and 98.48% of on-chain wealth is controlled by the top 1% of holders. Single-asset portfolios account for 73.50% of all accounts, and only 1.44% hold five or more assets. Multi-asset portfolios become more prevalent as account wealth increases, a pattern inconsistent with rational, risk-averse portfolio optimisation. Diversification is even lower than asset counts suggest: 81.0% of accounts keep at least 99% of their value in one asset. Portfolios are also largely static: in a typical month, 93.0% of accounts leave their holdings unchanged (Section 4).

(3) **Deviation from efficient frontier.** We measure the ℓ_1 weight distance between actual portfolios and three frontier strategies (minimum variance, maximum return, maximum Sharpe ratio). Distance rises with portfolio size: two-asset portfolios sit at or near the minimum-variance and maximum-return optima by construction, while those with five or more assets have mean distances of 48.7%–68.7%, depending on the strategy (Section 5).

[1] https://anonymous.4open.science/r/modern-portfolio-theory-in-the-wild-7F81/

(4) **Deviation effects.** We compare realised portfolio returns against MPT-optimal counterfactuals. Proximity to the constrained efficient frontier does not predict better risk-adjusted performance. Instead, about half of the variation in returns comes from market-wide movements that neither past market conditions nor allocation predict out of sample (Section 6).

The results have implications for both theory and practice. For theory, pervasive under-diversification and the dominance of market-wide movements over frontier proximity challenge the descriptive adequacy of MPT in the cryptoasset domain. The mismatch points to behavioural factors and market-wide movements, rather than mean-variance optimisation, as the organising principles of actual cryptoasset portfolios and their returns.

For practice, our findings caution against importing mean-variance frameworks wholesale into cryptoasset portfolio construction: rebalancing toward MPT optima does not reliably improve risk-adjusted performance, and passive market-capitalisation weighting offers a more robust default. The persistence of single-asset portfolios and the documented wealth concentration further suggest that frictions, behavioural factors, and investor-protection concerns deserve greater attention than allocation theory in shaping interventions across retail, advisory, platform, and regulatory contexts.

The remainder of the paper reviews related work, presents the method and findings for each objective, and concludes with a discussion of implications and limitations.

2 Background and Related Work

This section reviews the MPT framework and efficient frontier, empirical portfolio studies in traditional finance, organised around our three objectives, empirical studies in blockchain markets, and closes with the synthesis and research gap that motivates the present work.

2.1 Modern Portfolio Theory

The MPT framework was introduced by Markowitz [37], who proposed a method for allocating assets to maximise expected returns for a given level of risk, known as mean-variance optimisation. MPT relies on several assumptions: investors are perfectly rational and risk-averse; markets are frictionless (no transaction costs, taxes, or borrowing constraints); and prices are efficient, instantaneously reflecting all available information. Building on MPT, Sharpe [47] introduced the capital asset pricing model (CAPM), which models an asset's expected return as a function of its systematic risk relative to the market.

Central to MPT is the concept of the *efficient frontier*, the upper boundary of the minimum-variance set in expected-return/standard-deviation space: portfolios on the frontier achieve the highest possible expected return for a given level of risk, while portfolios below it are suboptimal. Any actual portfolio can thus be evaluated against the frontier, quantifying the gap between what an account holds and what mean-variance optimisation prescribes. For a comprehensive introduction to MPT and relevant literature, see Elton et al. [18].

2.2 Empirical Studies in Traditional Finance

A large body of empirical research has studied the portfolios of investors in traditional assets. Campbell [10] establishes household finance as a field, surveying evidence that while many households invest effectively, a significant minority (typically poorer and less educated) make portfolio mistakes. The findings summarised below provide context for each of our objectives.

Composition and diversification. A central finding is that investors hold fewer assets than theory prescribes. Goetzmann and Kumar [22] analyse more than 40,000 equity accounts at a U.S. discount brokerage and find that the average investor holds only four stocks (median of three), with greater under-diversification among younger, low-income, and less-educated investors. This

concentration often takes the form of familiarity bias: French and Poterba [21] document strikingly high domestic ownership shares despite substantial diversification gains from international investing, and Huberman [26] shows that shareholders concentrate holdings in geographically familiar companies. Benartzi and Thaler [4] further show that 401(k) participants tend to divide contributions equally across available options, the so-called *1/n heuristic*, another departure from mean-variance optimisation; DeMiguel et al. [15] find that this naive rule is hard to beat by mean-variance strategies out-of-sample, while Hellum et al. [25] show that this gap closes once the number of assets exceeds the number of training observations. Together, these studies characterise actual portfolios as simpler and less diversified than MPT prescribes.

Deviation from optimal allocations. Quantifying how far actual portfolios lie from theoretical optima requires both a benchmark and a distance measure. Using comprehensive administrative data on all Swedish households, Calvet et al. [9] measure the welfare costs of investment mistakes: while most households actually outperform their domestic stock index through international diversification, a minority incurs substantial losses from under-diversification. Massa and Simonov [39] further show that Swedish investors concentrate holdings in stocks related to their employment income and geographic proximity rather than hedging against non-financial income risk, a systematic deviation from theoretically optimal allocations. At the fund level, Didier et al. [16] find that even mutual funds exhibit limited international diversification, with global funds potentially achieving 2.6–5.5% higher annual risk-adjusted returns by broadening their allocations. These studies typically rely on welfare proxies or return differentials rather than portfolio-level distance metrics, leaving open how directly observable allocations compare to the efficient frontier.

Performance consequences. Individual investors generally underperform passive benchmarks. Barber and Odean [2] find that among 66,465 household accounts, the most active traders earned 11.4% annually versus 17.9% for the market, with overconfidence driving excessive trading. Odean [42] documents the disposition effect: investors realise gains more readily than losses, reducing annual returns by 4.4 percentage points. Classically, Brinson et al. [8] show that asset allocation policy explains 93.6% of the variation in quarterly portfolio returns across 91 large U.S. pension funds, with security selection and market timing playing minor roles, a benchmark against which we compare our findings in Section 7.

2.3 Empirical Studies in Blockchain Markets

Blockchain ledgers provide a uniquely complete empirical record of asset holdings; this subsection describes the Ethereum setting that makes portfolio reconstruction possible and reviews prior measurement work on token distributions and crypto-portfolio optimisation.

Ethereum as an empirical setting. Ethereum distinguishes two account types: externally owned accounts (EOAs), controlled by private keys, hold assets directly; contract accounts (CAs) implement programmable logic and hold assets on behalf of protocols such as decentralised finance (DeFi) applications for lending, trading, and liquidity provision. Two asset families live on Ethereum, and they differ in where their ownership records reside. Native ETH is the protocol's built-in asset: every account has an ETH balance in the protocol state, and ETH pays transaction fees and is issued as block and staking rewards. All other assets are tokens, managed by smart contracts that track each account's balance. Fungible tokens follow the ERC-20 standard, and wrapped assets such as wrapped ETH (wETH) and wrapped BTC (WBTC) are tokens redeemable one-to-one for an underlying asset [20]. The two asset families differ in observability. Token transfers are recorded as events in the blockchain logs, whereas ETH balances change through internal calls, fees, rewards, and withdrawals that appear only in execution traces, receipts, and block data. Both sources are public, which enables complete portfolio reconstruction (Section 3).

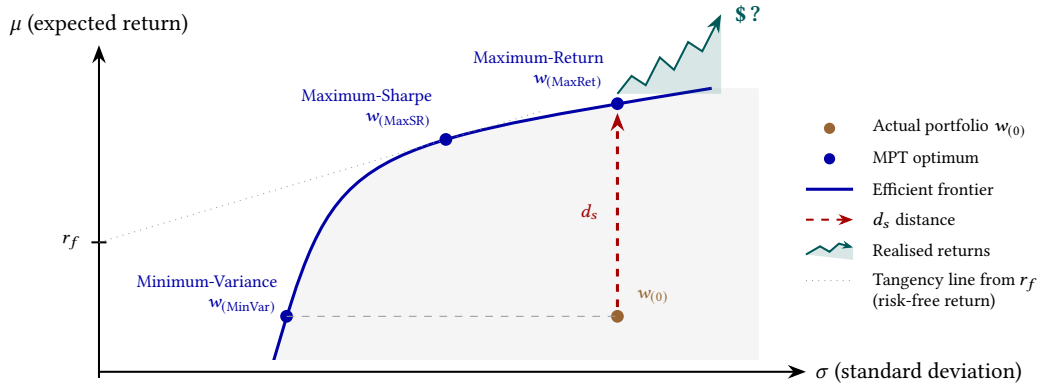


Fig. 1. Overall analytical approach in expected-return/standard-deviation space. Each actual portfolio ● $w_{(0)}$ is characterised (**Obj A**), compared to three frontier optima ● on the efficient frontier — via the weight-space distance - → d_s (**Obj B**), and evaluated on realised returns 〰 (**Obj C**).

On-chain measurement studies. Prior work has leveraged this data to characterise token distributions and network structure. Nadler and Schär [41] analyse the token distribution of DeFi governance tokens and find that protocol governance is heavily centralised. Victor and Lüders [50] conduct an early empirical study of Ethereum ERC-20 tokens over the first 6 million blocks, analysing token distributions and trade networks. These studies establish that on-chain holdings are concentrated, but analyse tokens individually and stop short of reconstructing complete portfolios at the account level or evaluating them against a portfolio-theoretic benchmark.

MPT applied to cryptoassets. A separate strand applies MPT to cryptoassets, typically on curated baskets rather than actual holdings. Brière et al. [7] examine Bitcoin's role in diversified portfolios; Platanakis and Urquhart [45], Trimborn et al. [48], and Petukhina et al. [44] apply mean-variance optimisation to small baskets of major cryptoassets. Borri et al. [6] apply an empirical asset pricing approach to tokens and find that their risk-adjusted performance is comparable to traditional markets. None of these studies compares actual account holdings to MPT-optimal allocations at population scale.

2.4 Synthesis and Research Gap

Together, these three strands establish that actual investors in traditional markets systematically deviate from MPT prescriptions, that complete on-chain cryptoasset portfolios can be measured at scale from public blockchain data, and that prior work on crypto-portfolio optimisation has focused on hypothetical baskets rather than actual holdings. Yet no study has characterised actual on-chain portfolio composition at population scale, measured how these portfolios compare to MPT-optimal allocations, or tested whether deviating from the efficient frontier has measurable performance consequences. Blockchain data enables this analysis at a completeness and scale impossible with traditional brokerage data; the present work fills this gap.

3 Data and Methods

3.1 Overall Approach

Our analysis reconstructs actual on-chain portfolios and evaluates them against MPT-optimal allocations along the three objectives. Figure 1 provides a schematic overview.

Each actual portfolio is represented by its asset-weight vector $w_{(0)}$, corresponding to expected return $\mu_{(0)}$ and standard deviation $\sigma_{(0)}$. We place $w_{(0)}$ in the feasible (μ, σ) region and characterise its composition and diversification (Obj A). We compute its weight-space distance d_s to three reference portfolios on the efficient frontier: the minimum-variance (MinVar), maximum-return (MaxRet), and maximum-Sharpe-ratio (MaxSR) optima. This quantifies deviation from MPT prescriptions (Obj B). We compare the realised returns of $w_{(0)}$ against those references over matched holding periods, testing whether proximity to the frontier predicts risk-adjusted performance (Obj C).

3.2 Data Collection

We collect data from two sources: the Ethereum blockchain, which provides event logs, execution traces, transaction receipts, and block data, and the Coingecko platform [12], a standard source of asset metadata and prices in prior studies [6, 17].

- **Ethereum blockchain:** We run an Erigon archive node [19] and export its full history from the genesis block (30 July 2015, block #0) through early December 2025. Smart contracts emit information to the blockchain *logs* when on-chain events occur; in particular, `Transfer` events following the ERC-20 standard record token movements between accounts and form the empirical basis for reconstructing token holdings. Native ETH movements are recorded in the execution *traces* (the full call tree of every transaction, including internal calls) rather than the logs (Section 2.3), so we also export these traces, together with transaction receipts and block headers and bodies, from which native ETH balances are reconstructed (Section 3.3). The log export ends at block #23,943,069 (5 December 2025) and the trace export at block #23,918,198; both lie beyond the last monthly snapshot (1 December 2025).
- **Coingecko:** We collect the complete Coingecko coin list with platform metadata and retain every coin that has an Ethereum contract address, which yields 5394 initial tokens with their contract address, name, and symbol. For native ETH and each token, we collect daily USD closing prices, market capitalisation, fully diluted valuation (FDV = Price × Total Supply), total circulating supply, and trading volume.

Figure 2 summarises the multi-stage cleaning pipeline described below. Of the 5394 initial tokens, 4562 (≈84.6%) pass all stages. Together with native ETH, which requires no cleaning, they form the asset universe of our analysis.

(1) **Standard compatibility.** We verify ERC-20 compliance by extracting the runtime bytecode from `CREATE` and `CREATE2` traces and checking for ERC-20 function and event selectors [50]. Proxy contracts are resolved to their underlying implementation.
(2) **Market activity.** We apply three filters for sufficient trading activity and reliable price data. First, we exclude tokens with fewer than 15 days of price history, which is insufficient for return estimation. Second, we exclude tokens with negligible historical trading volume. Third, we validate supply data, excluding tokens whose market capitalisation or FDV exceeds Ethereum's (an indicator of misconfigured supply); tokens for which Coingecko reports no supply data are retained.
(3) **On-chain consistency.** Finally, we validate that our event-based reconstruction matches the on-chain state. For each candidate token, we reconstruct balances at randomly sampled (account, block) pairs and compare them to the on-chain balance retrieved via RPC. Tokens with any disagreement between reconstructed and on-chain balances, or without a validated ledger, are excluded, since discrepancies indicate non-standard token implementations (e.g., rebasing tokens, tokens with transfer hooks, or tokens that modify balances outside `Transfer` events).

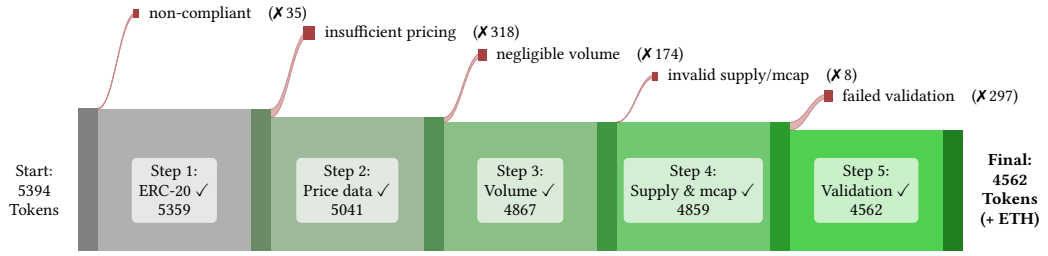


Fig. 2. Token cleaning pipeline. Out of 5394 initial tokens, 4562 pass all filtering stages (ERC-20 compliance; price history, trading volume, and supply/market-cap validation; reconstruction validation) and, together with native ETH, form the asset universe for subsequent analysis.

3.3 Portfolio Reconstruction

A cryptoasset portfolio comprises native ETH and ERC-20 tokens, while an archive node returns their exact balances via `eth_getBalance` and `balanceOf` only for one address, one asset, and one block per query, which does not scale to millions of accounts and thousands of assets across all snapshots. We therefore reconstruct balances from the recorded movements, by different routes for the two asset types. For native ETH, movements must be recovered from execution traces and protocol rules, while every ERC-20 token movement is announced as a `Transfer` event in the log stream. Both routes follow the same schematic steps (Figure 3): extracting every balance-changing event as a *transfer-like event* (Step 1), aggregating them into per-address ledgers of signed entries from which the balance at any block height is a prefix sum (Step 2), and pricing the balances into portfolio snapshots at target time points (Step 3). Token amounts and wei are integers, so every reconstructed balance is exact and reproduces these queries to the smallest unit.

3.3.1 Native ETH ledger. We reconstruct ETH balances from the complete history of balance changes: each change becomes a transfer-like event, and an address's balance at any block height is the sum of its signed entries up to that height.

Step 1: Event extraction. An ETH transaction is characterised by a sender, a recipient, and the value moved, as are the internal calls it triggers and the fee tip it pays to the block producer. Other balance changes have only one side: minted value, such as block rewards, staking withdrawals, and the genesis allocation, has only a recipient, and burned fees only a sender, so at every block the sum of all balances equals cumulative minted minus cumulative burned value. We convert all of them into transfer-like events, i.e., (sender, recipient, value) tuples, read from execution traces, transaction receipts, block data, and the genesis state; reverted calls are excluded, but their fees are kept. Appendix E.1 details all event classes and their sources.

Step 2: Address-level ledger. Each transfer-like event produces a debit entry for its sender and a credit entry for its recipient, or a single entry for minted or burned value. Because an address can receive and re-spend funds within one block, entries are sorted by address, block height, and intra-block index. The resulting ledger holds 7.43 billion transfer-like events from genesis to December 2025 as 12.9 billion signed entries, from which the balances of 279 million addresses are read as prefix sums. Further details on the ETH route are given in Appendix E.

3.3.2 ERC-20 token ledgers. **Step 1: Event extraction.** We extract the logs for the ERC-20 `Transfer` signature topic and write per-token files of transfer-like events, i.e., (sender, recipient, value) tuples, parallelised across tokens via a MapReduce pipeline [13]. Some tokens change

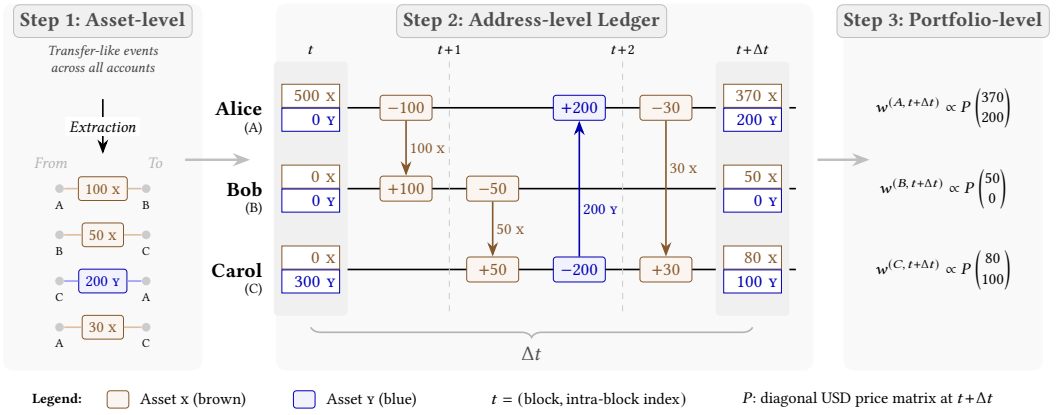


Fig. 3. Three-step reconstruction pipeline for two example assets (X in brown, Y in blue), each an ERC-20 token or native ETH, and three example accounts. **Step 1**: transfer-like events (`Transfer` logs for tokens, every balance-changing event for ETH) are extracted per asset. **Step 2**: each transfer-like event becomes a debit entry for the sender and a credit entry for the recipient, and cumulative sums over Δt (one month) give the balances at the next snapshot. **Step 3**: the balances are priced and normalised into portfolio weights.

balances outside `Transfer` logs: wETH, for instance, mints on `Deposit` and burns on `Withdrawal`, which we convert into transfer-like events from or to the zero address.

Step 2: Address-level ledger. As for ETH, each transfer-like event produces a debit entry for its sender and a credit entry for its recipient, here in per-(address, token) ledgers sorted by block height and log index. The ledger supports balance aggregation over any block range from genesis to a target height.

3.3.3 Portfolio snapshots. We treat each Ethereum address as an account, and a portfolio is the collection of cryptoassets held by an account at a given point in time.

Step 3: Portfolio-level aggregation. In this step, shared by both routes, we sum the corresponding entries of the token and ETH ledgers for each account and target timestamp, price each position with forward-filled daily USD closes, and normalise the position values into a fully-invested weight vector. Portfolios are computed for the *account universe*, every address that has ever sent or received one of the 4562 tokens (116 182 155 addresses). Addresses that have only ever moved ETH lie outside it, although their transfer-like events remain part of the ETH ledger. We sample *monthly snapshots* at the first day of each calendar month, matching the literature [22] and balancing resolution against compute, and observe 72 of them from January 2020 to December 2025, when the token ecosystem, its price coverage, and the account base are mature enough for population-scale analysis. Because ETH and wETH are redeemable one-for-one, we treat them as a single asset priced with the ETH series.

For each account a and time t, we obtain the USD value of each held position $v_i^{(a,t)}$ for asset i, portfolio value $V^{(a,t)} = \sum_i v_i^{(a,t)}$, number of held assets $N^{(a,t)}$, and fully-invested weight vector[2]

$$w^{(a,t)} = \left(w_1^{(a,t)}, \ldots, w_N^{(a,t)}\right), \; w_i^{(a,t)} = \frac{v_i^{(a,t)}}{V^{(a,t)}}, \; \sum_{i=1}^{N} w_i^{(a,t)} = 1. \tag{1}$$

[2]When (a, t) indices are suppressed (e.g., w, V, N, μ, σ), the quantity refers to account a's portfolio at time t.

We validate the reconstructed balances of both ERC-20 tokens and native ETH at randomly sampled (account, block) pairs against an Ethereum archive node.

3.4 Efficient Frontier Computation

Building on the MPT framework introduced in Section 2.1, we estimate the efficient frontier for each account and snapshot date t from empirical return statistics over a historical lookback window, constrained to the account's already-held assets. Asset-specific functionalities (e.g., underlying lending positions, staking yields) are disregarded: all cryptoassets are treated as simple assets characterised by prices.

3.4.1 *Return and risk estimation.*

Definition 3.1 (Expected return). The expected asset-return of asset i over a $T^{(-)}$-day lookback window ending at t is $\mu_i^{(t)} = \frac{1}{T^{(-)}} \sum_{\tau=0}^{T^{(-)}-1} r_i^{(t-\tau)}$ with $r_i^{(t)} = \log\left(\frac{p_i^{(t)}}{p_i^{(t-1)}}\right)$ the log-return of asset i at time t and $p_i^{(t)}$ the USD closing price of asset i at time t.

Definition 3.2 (Portfolio mean and variance). The expected return and variance of account a's portfolio at time t are $\mu^{(a,t)} = \sum_{i=1}^{N} w_i^{(a,t)} \mu_i^{(t)}$ and $\left(\sigma^{(a,t)}\right)^2 = \left(w^{(a,t)}\right)^\top \Sigma^{(t)} w^{(a,t)}$, where $\Sigma^{(t)} \in \mathbb{R}^{N\times N}$ is the empirical covariance matrix over the lookback ending at t, and N is the number of assets held at t.

3.4.2 *Reference portfolios under optimisation strategies.*

For each account a and time t, we apply three optimisation strategies, $s \in \{\text{MinVar}, \text{MaxRet}, \text{MaxSR}\}$, each producing a reference portfolio $w_{(s)}$ on the constrained efficient frontier.

Definition 3.3 (Frontier portfolios). Let $w_{(0)} \equiv w^{(a,t)}$ denote the actual portfolio with moments $\mu_{(0)}, \sigma_{(0)}$, and $w_{(s)}$ the portfolio optimised under strategy s with moments $\mu_{(s)}, \sigma_{(s)}$:

Min variance at target return (MinVar): $\min_w \sigma_{(s)}$ s.t. $\mu_{(s)} = \mu_{(0)} \longrightarrow w_{(\text{MinVar})}$.
Max return at target volatility (MaxRet): $\max_w \mu_{(s)}$ s.t. $\sigma_{(s)} = \sigma_{(0)} \longrightarrow w_{(\text{MaxRet})}$.
Max Sharpe ratio (MaxSR): $\max_w (\mu_{(s)} - r_f)/\sigma_{(s)} \longrightarrow w_{(\text{MaxSR})}$.

The MinVar and MaxRet anchor to the actual portfolio's expected return and standard deviation, respectively, corresponding to the horizontal and vertical projections of $w_{(0)}$ onto the frontier (Figure 1). The MaxSR is the *tangency portfolio*, the point where the line from r_f touches the frontier, and it depends on the assets held but not on their initial weights. We set $r_f = 0$ for the MaxSR optimisation, the same convention used for the CAPM alphas in Section 3.5: no genuinely risk-free on-chain asset exists over the study period, and a zero rate avoids importing an off-chain yield assumption into an on-chain analysis. Section C.2 motivates the natural alternative calibration of 5% per annum, anchored in the U.S. Treasury-bill convention of [45] and matching on-chain stablecoin lending yields (2023–2024 [5]); re-running the MaxSR optimisation at this rate leaves the distances essentially unchanged.

3.4.3 *Shared constraints.*

All three optimisation problems share the following constraints, which with the held-asset restriction define the *constrained efficient frontier* used throughout:

- *Fully invested*: $\sum_i w_{(s),i} = 1$, with $w_{(s),i} = 0$ whenever $w_{(0),i} = 0$ (pure reallocation across held assets, no external funds).
- *Long-only*: $w_{(s),i} \geq 0$ (no spot-level short-selling available for most cryptoassets).
- *Single-asset cap*: $w_{(s),i} \leq w_{\max} = 0.9$ (prevents collapse to a near-single-asset solution).

Before optimisation, the actual allocation $w_{(0)}$ is itself projected onto this constraint set, so that the anchors $\mu_{(0)}$ and $\sigma_{(0)}$ of the MinVar and MaxRet problems are attainable on the constrained

frontier. The distances of Section 3.5 are measured from the projected allocation, whereas realised returns are computed from the actual holdings.

3.4.4 Configuration. We solve all three optimisation problems with sequential least-squares programming (SLSQP), supplying analytic gradients for all objectives and constraints.[3] The following parameters are shared across all optimisation instances:

- *Lookback window:* $T^{(-)} = 60$ daily observations before the snapshot, with prices forward-filled; an account is optimised only if its held assets share at least 45 days of prices in the window. The window balances stable covariance estimation against the frequent regime changes of cryptoasset markets.
- *Covariance shrinkage:* We apply Ledoit–Wolf shrinkage [32] to stabilise the empirical covariance matrix when N is large relative to $T^{(-)}$.
- *Return regularisation:* Sample mean returns are highly noisy over the $T^{(-)}$-day lookback in crypto markets. We therefore estimate each asset's expected return by a 5% trimmed mean of its daily log returns, μ_i^{trim}, and shrink it toward a neutral prior of equal expected returns across assets: $\mu_i^{\text{shrunk}} = \lambda\,\bar{\mu} + (1-\lambda)\,\mu_i^{\text{trim}}$, where $\bar{\mu} = \frac{1}{N}\sum_{j=1}^{N}\mu_j^{\text{trim}}$ is the cross-asset mean and $\lambda = 0.5$ (further details in Section C.1).

3.5 Distance and Performance Metrics

We evaluate the optimisation outcomes along two dimensions: how far each optimised portfolio deviates from the actual allocation (Obj B), and whether this improves risk-adjusted returns (Obj C).

3.5.1 Weight ℓ_1 distance.

Definition 3.4 (Weight ℓ_1 distance). The ℓ_1 distance between the actual allocation $w_{(0)}$ and an optimised reference $w_{(s)}$ is $d_s = \frac{1}{2}\left\|w_{(0)} - w_{(s)}\right\|_1 = \frac{1}{2}\sum_{i=1}^{N}\left|\,w_{(0),i} - w_{(s),i}\,\right|$.

As derived in Section C.1, we use ℓ_1 rather than ℓ_2 as d_s admits a financial interpretation: it equals the minimum one-way turnover to rebalance from $w_{(0)}$ to $w_{(s)}$, and is bounded in $[0,1]$ for long-only, fully-invested portfolios, with 0 indicating identical allocations and 1 disjoint ones.

3.5.2 Risk-adjusted performance.

Definition 3.5 (Forward-realised return). The forward-realised return of an allocation w of account a, whether the actual $w_{(0)}$ or any theoretical $w_{(s)}$, between consecutive snapshots t and $t' = t + T^{(+)}$ is $r_{t'}^{(a,t)} = \sum_{i=1}^{N} w_i \left(\frac{p_i^{(t')}}{p_i^{(t)}} - 1\right)$, where $p_i^{(t)}$ and $p_i^{(t')}$ are the asset prices at t and t'.

Raw returns conflate allocation with market movement: in a bull market, even naive allocations are positive. To isolate the allocation's contribution, we apply the capital asset pricing model (CAPM) and decompose the return into a component proportional to the portfolio's market exposure (β) and an excess residual (α), the risk-adjusted return. As the market benchmark $r_{(m)}^{(t)}$, we take the equally-weighted average of the daily log-returns of wETH and WBTC. Equal weighting prevents structural overlap with the market-cap-weight strategy. Daily log-returns enter the β estimation below; over the forward window from t to t', the market return $r_{(m)}^{(t')}$ is the equally-weighted *simple* holding-period return of the two benchmarks, matching the convention of Definition 3.5.

[3] The two-asset MaxRet case is solved in closed form: with $w = (x, 1-x)$, the variance-equality constraint is a quadratic in x, and the optimum is the in-bounds root with the higher expected return. If the numerical solver fails to converge on a MaxRet instance, the pipeline substitutes the tangency solution for that account; if a MinVar instance fails, the return-equality constraint is first relaxed to $\mu_{(s)} \geq \mu_{(0)}$ and, failing that, the global minimum-variance portfolio is substituted. Both substitutions are flagged in the output.

Definition 3.6 (Market exposure). The market exposure of account a's portfolio at time t is $\beta^{(a,t)} = \sum_i w_i^{(a,t)} \beta_i^{(t)}$, with $\beta_i^{(t)} = \frac{\mathrm{Cov}_\tau\left(r_i^{(t-\tau)}, r_{(m)}^{(t-\tau)}\right)}{\mathrm{Var}_\tau\left(r_{(m)}^{(t-\tau)}\right)}$ the sensitivity of asset i to the market at time t, estimated over the same $T^{(-)}$-day lookback, and $w_{(s)}$ admissible in place of $w^{(a,t)}$.

A value $\beta^{(a,t)} = 1$ implies lockstep movement with the market, $\beta^{(a,t)} > 1$ amplifies market moves, and $\beta^{(a,t)} < 1$ dampens them.

Definition 3.7 (Risk-adjusted excess return). For account a's allocation held from t to t', the part of the forward-realised return not explained by market exposure is: $\alpha_{t'}^{(a,t)} = r_{t'}^{(a,t)} - \beta^{(a,t)} \cdot r_{(m)}^{(t')}$.

Here, $r_f = 0$ as everywhere in the pipeline (Section 3.4). Positive $\alpha_{t'}^{(a,t)}$ indicates outperformance beyond market exposure, and negative values indicate underperformance.

Ethics and data handling. The study uses public Ethereum logs and execution traces, public price data, and aggregate address tags. We do not deanonymise addresses, and address tags are used only for aggregate contract/category-level analysis. The released artifacts do not expose private address-level information, only public data already publicly available on-chain.

Use of generative AI. We used Anthropic's Claude models (via the Claude web interface and Claude Code) for language editing and for code assistance in data processing and plotting. AI was not used to generate results, analyses, or scientific claims. All AI-assisted output was reviewed and verified by the authors, who take full responsibility for the content of this paper.

4 Portfolio Characteristics

We now address **Obj A** by characterising the wealth, concentration, diversification, and trading activity of on-chain portfolios, covering the 116 182 155 (116 M) accounts that have ever held one of the 4562 tokens, at 72 monthly snapshots from January 2020 to December 2025 (Section 3.3).

4.1 Wealth and Its Concentration

We measure account wealth V as the USD value of assets held at each monthly snapshot. Total wealth follows the well-known boom–bust–recovery cycle of the broader crypto market and of Ethereum total value locked (TVL) in DeFi protocols [14] (Appendix B.1). Disaggregating wealth by account type reveals an asymmetry: while CAs make up on average only 1.33% of the accounts holding assets, they hold 26.26% of total wealth, while EOAs account for the remaining 73.74%. TVL captures only value locked in DeFi protocols and thus misses the large share held in EOA accounts.

We characterise the wealth distribution across accounts from both ends: first via shares of accounts by wealth bin and account type, then via the top 1% and 10% wealth shares as our primary measure of concentration, complemented by standard inequality indices.

As Figure 4 shows, the distribution of accounts across wealth bins differs systematically by account type but converges on the same headline: the majority of both CAs and EOAs consistently hold less than \$100, with CAs concentrated in the dust[4] bin ($\leq$ \$1, 47.52% share over the entire observation period) and EOAs in the \$1–\$100 bin (53.45%). The two types reach their low-wealth-share minima at different times, the CA minimum in May 2021 at the top of the first 2021 bull run and the EOA minimum in December 2021 at the broader market peak, jointly producing the two wealth peaks of 2021. Account growth trajectories, CA/EOA wealth shares over time, and the full bin-level distributions are reported in Appendix B.1.

[4]Millions of accounts can hold negligible residual balances after a trade, which can inflate the denominator of the distribution and mechanically increase measured concentration.

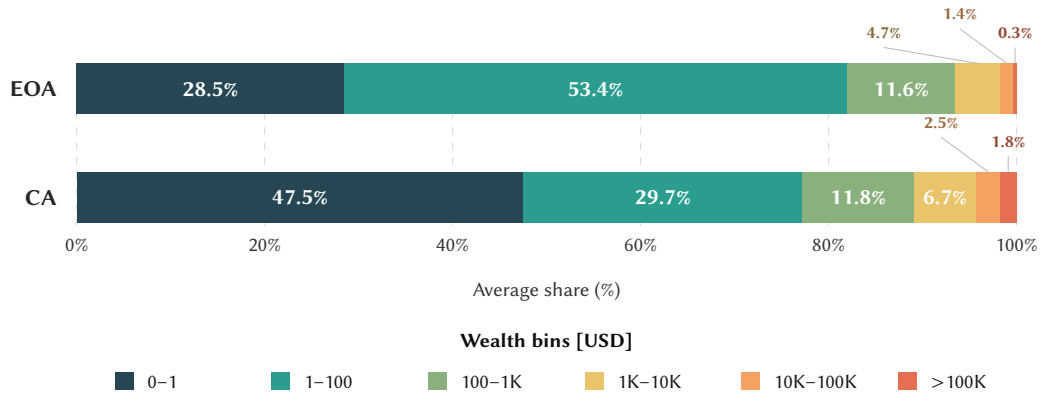


Fig. 4. Share of EOA and CA accounts by wealth bucket (USD), computed across all 72 monthly snapshots.

Turning to the top of the distribution, we restrict the analysis to accounts whose portfolio value exceeds $1, to mitigate the "dust" problem noted above.

We find that the top 1% of accounts consistently controls the vast majority of total portfolio value, increasing from 95.88% to 98.48% over 2020–2025. The top 10% likewise holds nearly the entire ecosystem value, always above 99.2%, while the number of accounts grows almost sixfold over the same period. These findings are corroborated by standard inequality measures (Gini, HHI), which remain uniformly high throughout the observation period (Appendix B.4).

Narrowing to the top 0.1%, account type composition departs from the ecosystem average: CAs make up 11.90% of these addresses, about nine times their overall 1.33% share, with DeFi protocols, exchanges, and custodial wallets dominating the tagged categories. However, the aggregate CA value share within this slice (28.17%) is essentially unchanged from the ecosystem-wide 26.26%, meaning the same fraction of the pie is simply held by a proportionally larger group. The EOAs that reach the top 0.1% are correspondingly whales with highly concentrated individual wealth.

Yearly concentration statistics and account type breakdowns are provided in Appendix B, as well as the full Gini and HHI analysis and tag-based attribution details.

4.2 Portfolio Size

Having characterised wealth levels and concentration, we now examine portfolio size N, the number of distinct assets held, as a measure of diversification. We focus on accounts holding at least one asset at the snapshot. Because the account universe is defined by token activity (Section 3.3), a single-asset portfolio is either one token or the native ETH balance of an account that holds no token at that snapshot. Addresses that never held a token are not part of the population.

Figure 5 shows the distribution of portfolio sizes across wealth categories, averaged over monthly snapshots. Overall, 73.50% of accounts hold only a single asset, 20.35% hold two, 4.70% hold three to four, and only 1.44% hold five or more distinct assets. Single-asset accounts remain the largest group in every wealth category, but their share falls from 90.9% in the lowest category to 32.7% in the highest, so the share of multi-asset portfolios rises with wealth and flattens above $1,000. Diversification is shallow even where it occurs: two-asset portfolios make up 76.5% of all multi-asset portfolios, and CAs are less diversified than EOAs, with 84.08% versus 73.36% holding a single asset.

Counting assets overstates diversification when weights are highly unequal. We therefore also measure each account's largest portfolio weight and its effective number of assets, $N_{\text{eff}} = 1/\sum_i w_i^2$, the inverse of the Herfindahl-Hirschman index (HHI) of its portfolio weights w_i, which equals N

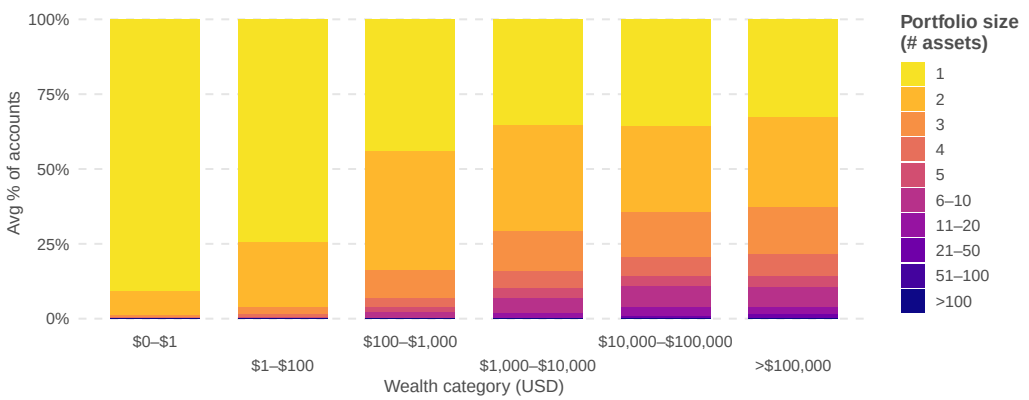


Fig. 5. Portfolio size distribution: share of accounts holding a given number of distinct assets, by wealth category, averaged across all monthly snapshots.

only for equal weights. On average, 81.0% of accounts keep at least 99% of their value in a single asset, and only 1.9% reach $N_{\text{eff}} \geq 2$, that is, are at least as diversified as an equal split between two assets. The median multi-asset portfolio keeps 91.8% of its value in its largest position. The count N is further inflated by dust: ignoring positions worth less than \$1 reduces the number of multi-asset portfolios by 37.7% and that of portfolios with five or more assets by 39.2%, while the concentration measures remain unchanged. Wealthier accounts hold more assets but not more balanced portfolios: multi-asset accounts above \$100,000 hold 4.6 assets on average, yet their median largest weight is 99.5%, compared with 88.6% in the least concentrated category, \$100–\$1,000 (Appendix B.5).

4.3 Trading Activity

Snapshots show where portfolios stand rather than how often they change. For every pair of consecutive snapshots, we compare an account's actual weights with the weights its earlier holdings would have after price changes alone. Half the ℓ_1 difference between the two, the turnover, is the share of the portfolio that was actively reallocated (Appendix B.6). In a typical month, 93.0% of positive balance accounts do not change holdings (87.3% of multi-asset accounts). Only 5.9% of multi-asset accounts reallocate more than a quarter of their portfolio, and this minority accounts for at least 72% of the reallocation by multi-asset accounts. Trading rises with wealth: the share of multi-asset accounts without any change falls from 97.9% below \$1 to 59.2% above \$100,000. Within the same wealth categories, the share of accounts reallocating more than a quarter of their portfolio has roughly halved since 2021. Thus, portfolios are largely static: the distances in Section 5 describe where buy-and-hold portfolios sit rather than allocation choices renewed at every snapshot.

Takeaways

CAs account for only 1.33% of asset-holding addresses but hold 26.26% of aggregate wealth, so TVL systematically undercounts on-chain wealth by ignoring the EOA-held share. On-chain wealth is highly concentrated: the top 1% of accounts controls 98.48% in 2025. While 73.50% of accounts hold only a single asset, the multi-asset share rises with wealth and flattens above \$1,000. Diversification is even lower than asset counts suggest: 81.0% of accounts keep at least 99% of their value in one asset, and wealthier accounts add assets without balancing them. Portfolios are also largely static: in a typical month, 93.0% of accounts leave their holdings unchanged. *Overall, on-chain portfolios*

are small along both dimensions and rarely change: most accounts hold modest wealth (<$100) in only one or two assets, keep nearly all of it in one, and leave it untouched from month to month.

5 Distance to the Efficient Frontier

We now tackle **Obj B** by measuring how far portfolios sit from the constrained efficient frontier and identifying suboptimality predictors. For every account holding at least two assets, we solve the three mean-variance optimisation (MVO) problems by following strategies (MaxRet, MinVar, MaxSR) at 72 monthly snapshots (Jan 2020 – Dec 2025), yielding 817 525 786 (818 M) account-month optimisations. Since most portfolios do not change between snapshots (Section 4.3), these distances describe where largely static portfolios sit, not allocation choices renewed every month. We provide supplementary analyses in Appendix C and computational details in Appendix E.

5.1 Measuring Portfolio Distance

We quantify the gap between actual ($w_{(0)}$) and optimal ($w_{(s)}$) allocations using the weight distance d defined in Definition 3.4, examining overall means, distributional patterns across the three strategies, and temporal stability. We rescale the raw distance from $[0, 1]$ to $[0, 100]$% throughout, where 0% denotes coincidence with the optimum and 100% zero weight overlap.

Distance prevalences. Figure 6 groups the distances into buckets for each of the three optimisation strategies, reporting shares pooled over the entire observation period. The resulting distances reveal a clear ordering across strategies: on-chain portfolios are closest to the MinVar optimal solution (7.11% mean), followed by MaxRet (12.24%), and farthest from the MaxSR optimum (41.86%). However, these means mask important distributional differences: the three strategies show strikingly different profiles. MinVar and MaxRet concentrate the bulk of account-observations at the optimum (78.43% and 69.75% within 1%, respectively), with a thin tail toward large distances. In contrast, MaxSR places most observations far from the optimum: only 17.86% lie within 1%, while 27.45% fall in the $(60, 80]$% bin, the largest of all bins.

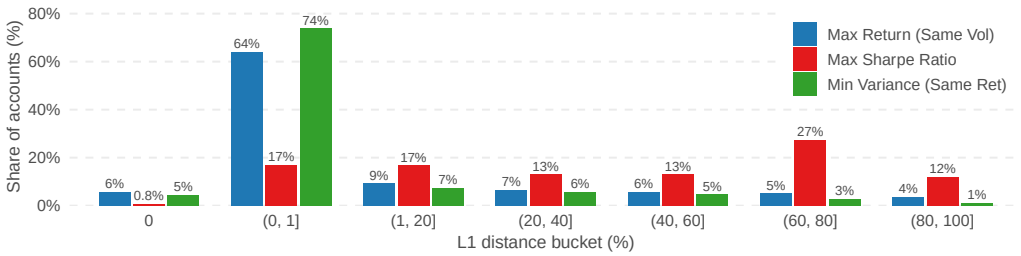


Fig. 6. Distribution of portfolio distance to the constrained efficient frontier for the three optimisation strategies. Share of 818 M account-month optimisations across seven distance bins.

Temporal dynamics. Distance may vary with portfolio size and evolve with market conditions over time. To investigate, we stratify accounts at an illustrative threshold of $N = 5$ for portfolio size and track the mean distance over time (Figure 23 in Appendix C.6). All three distances exhibit a slight upward trend over time, most visibly for large portfolios.

Small portfolios ($N < 5$) have a consistently small average distance to MinVar (4.8 %) and MaxRet (9.8 %), indicating near-structural alignment, while MaxSR shows the largest (40.3 %). Large portfolios ($N \geq 5$) show a sharp increase to 48.7 % for MinVar, 55.4 % for MaxRet, and 68.7 % for MaxSR. A paired Wilcoxon signed-rank test across the 72 monthly snapshots confirms this small–large gap is significant with large effect sizes.

Notably, this near-optimal share is almost entirely due to two-asset portfolios, which have little room to deviate from the optimum and therefore sit close to it by construction: with two assets, the return constraint of MinVar admits only the account's own weights, and the volatility constraint of MaxRet at most one alternative. Once we restrict to $N \geq 5$, the share within 1% of the optimum drops to near zero. Test statistics, Cohen's d values, per-size-bucket plots, threshold sensitivity, cross-strategy correlations, and robustness analysis are in Appendix C.

5.2 Determinants and Modeling of Distance

Having characterised how distance is distributed and evolves over time, we now ask what predicts suboptimality and whether it admits a parsimonious model. We first identify the dominant determinants, then formalise the distance–size relationship ($d \sim N$) with a parametric model, and finally benchmark actual allocations against naive strategies.

On a sampled subset, we regress distance (d) on portfolio size (N), the log portfolio value ($\log V$), and snapshot month (t). As shown in Figure 7, N is the dominant determinant of *mean* distance across strategies. Random Forest models show that N also dominates the variation of MinVar and MaxRet distance across individual accounts (95.3% and 78.7% of feature importance), driven by a near-optimal cluster of almost exclusively two-asset portfolios, while for MaxSR importance is split almost evenly between portfolio value (35.7%), the snapshot month (33.8%), and N (30.5%).

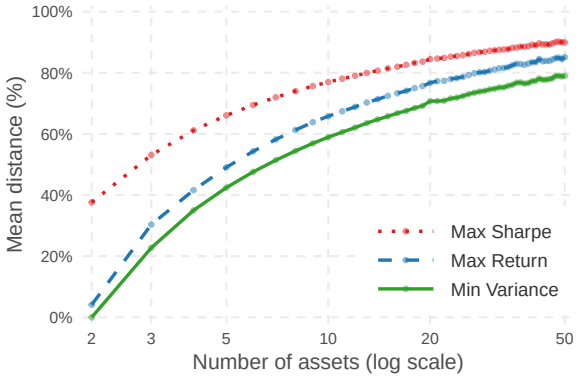


Fig. 7. Mean distance vs. portfolio size. Distance d as a function of size N on a log-scaled axis for all depicted strategies.

Table 1. Power-decay distance model. Fitted parameters of Equation (2) by strategy.

Strategy s	ψ_s	γ_s	δ_s^{∞} (%)
MaxRet	1.9014	1.0033	80.60
MinVar	1.7446	0.8030	81.66
MaxSR	1.0491	0.8401	90.88

Parametric model of distance. As Figure 7 shows, the $d \sim N$ relationship is monotone and concave across all three strategies: distance rises steeply from $N = 2$ to $N = 5$ and flattens thereafter, consistent with diminishing marginal deviation as portfolio size grows. This flattening also motivates the $N \geq 5$ stratification used in Section 5.1. We therefore formalise it with a parsimonious three-parameter power-decay model:

$$\hat{d}_s(N) = \delta_s^{\infty}\left(1 - \psi_s \cdot N^{-\gamma_s}\right), \tag{2}$$

where $\hat{d}_s$ is the fitted mean distance in percent, $N \geq 2$ the portfolio size, $\delta_s^{\infty} \leq 100\%$ the asymptotic ceiling, $\psi_s > 0$ a shape parameter setting the depth of the dip from the asymptote, and $\gamma_s > 0$ the exponent governing how quickly distance saturates as N grows. Table 1 reports the fitted parameters, and all three fits achieve $R^2 \geq 0.99$. No asymptote reaches 100%: MinVar and MaxSR plateau at 81.66% and 90.88%, implying persistent structural overlap with the minimum-variance and Sharpe-optimal allocations, while the MaxRet fit is dominated by its steep rise at small N, and

its asymptote (80.60%) lies below the mean distance of the largest portfolios. Estimation details, goodness of fit, and fitted curves are in Appendix C.4.

Comparison with naive allocation benchmarks. To put these distance results into perspective, we compare the optimisation strategies against two naive allocation benchmarks that require no estimation of returns or covariances: an *equal-weight* portfolio ($w_i = 1/N$) and a *market-capitalisation-weight* portfolio. The mean distance to equal-weight is 36.0% and to market-capitalisation-weight 41.9%, both substantially larger than the 7.11% distance to MinVar. The medians reveal a distributional difference: for equal-weight the median (40.0%) exceeds the mean (36.0%), whereas for market-capitalisation-weighted (33.5% median, 41.9% mean) the reverse holds. The typical account is thus noticeably closer to market-capitalisation-weight than to equal-weight portfolios, which is intuitive: concentrated holdings in ETH or another large-cap asset mechanically resemble a market-cap-weighted portfolio, even without deliberately targeting the weighting.

This benchmark comparison examines the structural effect of MPT optimisation, revealing that the three strategies shift accounts in qualitatively distinct ways. MaxRet and MinVar both produce a negative mean distance-delta against equal-weight, meaning optimisation on average moves accounts *closer* to an equal-weight allocation; both additionally leave most accounts unchanged (71.0% and 78.2%), consistent with many portfolios already lying at the constrained optimum. MaxSR, by contrast, moves more accounts farther from both equal-weight and cap-weight than closer to them (full decomposition in Appendix C).

Takeaways

Across strategies, portfolios are closest to the minimum-variance optimum (MinVar, mean 7.11%) and farthest from the Sharpe-optimal frontier (MaxSR, 41.86%). The large near-MinVar share (distance $\leq$ 1%: 78.43%) is structural: two-asset portfolios coincide with the minimum-variance solution, and the share collapses for larger portfolios ($N \geq 5$). Portfolio size N is the dominant determinant of the mean distance to the constrained efficient frontier across all three strategies. Larger portfolios exhibit substantial distance to all three frontiers, indicating considerable room for rebalancing. Relative to naive benchmarks, the typical actual portfolio is nearer to market-cap-weighting than to equal-weighting, reflecting concentrated holdings in large-cap assets. MaxRet and MinVar optimisation moves accounts *closer* to equal-weight, whereas MaxSR on balance pushes them farther from both naive benchmarks. *Overall, portfolios sit closest to the minimum-variance frontier, but this proximity is largely explained by the dominant two-asset portfolio size.*

6 Realised Portfolio Performance

Finally, we address **Obj C** by quantifying how deviations from the constrained efficient frontier translate into realised performance relative to the market. For each account snapshot, we compute the $T^{(+)}$-day realised return under the actual, optimised, and naive allocations.

6.1 Forward Returns and Risk-Adjusted Returns

We set $T^{(+)} = 20$ days, keeping adjacent forward returns approximately non-overlapping [24] and yielding a 3:1 $T^{(-)}/T^{(+)}$ ratio standard in rolling mean–variance evaluation [15, 30]. The allocation $w_{(0)}$ serves as baseline against optimised MPT strategies (MaxRet, MinVar, MaxSR) and naive allocations (equal-weight, market-cap-weight). Table 2 reports, for the forward-realised return $r_{t'}^{(a,t)}$, the median and the hit rate, which is the fraction of accounts beating the baseline among those whose strategy portfolio differs from their own by more than 1% in weight distance (Section 5.1); closer portfolios count as ties. Alongside, the risk-adjusted return α is documented with the median and the fraction of accounts with positive results for each strategy. Throughout, we compute

medians and shares within each monthly snapshot and report their median across snapshots, so every snapshot carries equal weight and cohort-size growth does not dominate statistics.

Table 2. Realised returns and risk-adjusted performance by strategy. Median forward return over the $T^{(+)}$-day horizon, hit rate versus the baseline allocation, and CAPM α. Hit rates exclude ties, i.e. accounts whose strategy portfolio lies within 1% of their own (70.7% for MaxRet, 78.4% for MinVar, 18.6% for MaxSR).

Strategy	Median Return (%)	Hit Rate (%)	Median α (%)	Positive α (%)
Baseline (actual)	0.25	—	−0.64	34.3
MaxRet	0.26	51.2	−0.36	45.0
MinVar	0.41	52.8	−0.20	43.3
MaxSR	0.44	46.0	−0.36	45.7
Equal-weight ($1/N$)	0.36	49.7	−0.65	41.1
Market-cap-weight	1.01	60.6	−0.27	8.2
Market benchmark	1.94	—	—	—

While the market benchmark r_m reaches 1.94%, the baseline realised return is 0.25% for the actual allocation. Market-cap-weighting achieves the highest hit rate (60.6%) and the highest median return (1.01%). MinVar and MaxRet also exceed a 50% hit rate (52.8% and 51.2%), with equal-weight marginally below (49.7%) and MaxSR below (46.0%).

The risk-adjusted return is negative for every allocation. MinVar has the least negative median α (−0.20%), followed by market-cap-weighting (−0.27%); all three MPT strategies are marginally less negative than the baseline (α of −0.36%, −0.20%, and −0.36% vs. −0.64%), while equal-weighting is not (−0.65%). The positive-α fraction is highest for MaxSR (45.7%) and lowest for market-cap-weighting (8.2%), but the latter reflects ETH rather than allocation: 94.1% of multi-asset portfolios hold ETH, and cap-weighting concentrates nearly every portfolio on it, so that within a snapshot almost all or almost none of the cap-weighted portfolios earn a positive α (fewer than 10% in 37 and more than 90% in 32 of the 72 snapshots).

The allocation also changes little how often an account beats the market: in the median snapshot, the share of accounts beating the wETH–WBTC market index varies only from 38.1% to 42.6% across the actual, MPT, and equal-weight allocations, and the share with positive α from 34.3% to 45.7%, against swings from 3.32% to 90.44% across snapshots (Appendix D.1).

6.2 Predictors of Performance

To identify which account characteristics predict performance, we train Random Forest models on six features: snapshot month, portfolio value, portfolio size N, portfolio β, and binary indicators for holding ETH (native, wrapped, or liquid-staked) or BTC wrappers.

The snapshot month dominates both forward and risk-adjusted returns, accounting for 64–75% of the permutation importance for realised returns ($R^2 = 0.85$–0.88) and 44–52% for α ($R^2 = 0.74$–0.79). Portfolio β is the second strongest signal, modest for realised returns (17–20%) and more pronounced for α (28–31%), while portfolio value contributes less (6–15% and 12–25%) and size N and asset composition contribute marginally. This month importance, however, measures market-wide movements rather than predictable timing: the random train–test split places every month in both samples, so the models learn each month's average outcome. About half of the variance of 20-day returns lies between months (50.7–55.6%, against 18.2–19.4% for α), and this part is not predictable out of sample: with whole months held out of training, the same models reach $R^2 = -0.44$ for returns and −0.29 for α, and ex-ante market-condition features fare no better

($R^2 = -0.04$ and -0.03). Within a month, account characteristics, frontier distance included, explain none of the variation either ($R^2 = -0.09$ and -0.06).

Figure 8 shows the same pattern under repeated monthly optimisation: the cumulative excess returns over the market benchmark largely move together across allocations. After early 2022, every allocation loses ground to the market, the actual allocation falling from +60.1 pp in March 2022 to −35.5 pp; only market-cap-weighting, which tracks ETH, ends above the market (+39.3 pp).

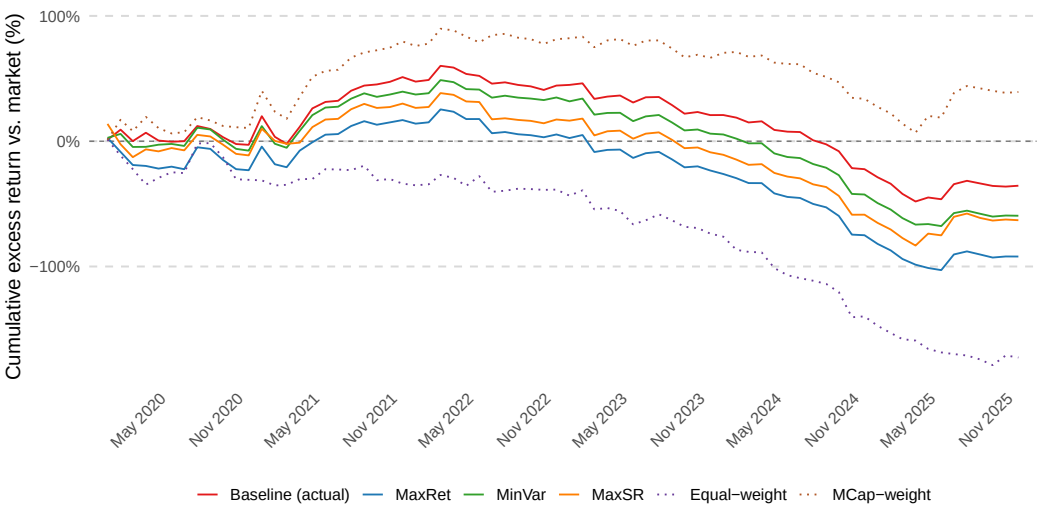


Fig. 8. Cumulative excess return over the market benchmark. Running sum of monthly median return differentials by strategy, with allocations re-optimised independently at each snapshot.

We furthermore provide Random Forest, SHAP, ℓ_1-distance–α, out-of-sample, and per-snapshot robustness checks in Appendices D.1, D.2, D.3, and D.4.

Takeaways

The market benchmark outperforms all optimisation strategies, including MPT, in median return, and the risk-adjusted return is negative for every allocation. Returns are driven by market-wide movements: about half of the variation in accounts' 20-day returns (50.7–55.6%) is shared by all accounts in the same month, and neither past market conditions nor account characteristics predict it out of sample, while frontier distance is unrelated to risk-adjusted returns. *Overall, the month in which returns are realised matters more than how an account allocates, yet neither the month's outcome nor the allocation's contribution can be predicted.*

7 Discussion

7.1 Key Findings

We present the first large-scale empirical study comparing actual on-chain portfolios to their theoretically optimal allocations under MPT. We propose a method for reconstructing complete cryptoasset portfolios and apply it to 116 M Ethereum accounts.

First, we find that wealth is highly skewed and portfolios are small: the top 1% of accounts control 98.48% of on-chain wealth in 2025, and 73.50% of accounts hold a single asset. This extends prior evidence on narrow sets of assets, such as governance tokens [3, 41, 43], to a much larger set that

includes native ETH. Single-asset dominance is far more pronounced than the typical US retail holding of three to four stocks [22], and diversification is lower still than asset counts suggest: 81.0% of accounts keep at least 99% of their value in one asset. We find little evidence of the naive $1/N$ heuristic [4]: the median multi-asset portfolio keeps 91.8% of its value in its largest position. Instead, portfolios are largely static, with 93.0% of accounts leaving their holdings unchanged in a typical month, much like the participants of a large 401(k) plan, whose allocations are mostly extreme and rarely reshuffled [1].

Second, we measure how far portfolios sit from the constrained efficient frontier. Small portfolios lie closest to the minimum-variance optimum, but this proximity is structural rather than deliberate: portfolios near the optimum are predominantly two-asset portfolios that mechanically coincide with the minimum-variance allocation. Apparent optimality erodes as portfolios grow; for five or more assets, mean deviations from the frontier reach 48.7%–68.7%, depending on the optimisation strategy. We further formalise the relationship between portfolio size and mean distance to the theoretical optima through a parsimonious three-parameter power-decay model.

Third, we quantify how deviations from the efficient frontier translate into realised performance relative to the market. Passive market-capitalisation weighting outperforms every MPT optimisation strategy in median realised return, yet the median risk-adjusted return is negative for every allocation, and frontier distance is unrelated to it. The dominant driver of realised returns is the month in which they are realised, not how an account weights the assets it holds: the snapshot month leads the permutation importance for returns (64–75%) and for risk-adjusted returns (44–52%), ahead of portfolio β, value, and size, and about half of the variance of returns (50.7–55.6%) lies between months. This month effect reflects market-wide movements rather than predictable timing: with whole months held out of training, neither account characteristics nor ex-ante market conditions predict it (Appendix D.4). Moreover, under repeated monthly optimisation, the cumulative excess returns of all allocations largely move together: after early 2022, every allocation loses ground to the market benchmark, and only market-cap weighting ends above it.

These results parallel established findings from traditional finance, which attribute roughly 90% of the time-series variation in portfolio returns to asset allocation policy [8, 27], a share that stems primarily from participation in the capital markets in general rather than from each fund's specific policy [27]. On-chain, too, market-wide movements dominate, and reweighting the held assets changes little. A plausible explanation is offered by Liu et al. [34], who show that a small number of common factors (in particular a cryptocurrency market factor) capture the variation in returns across cryptoassets, leaving limited scope for diversification benefits across tokens. On Ethereum, this factor has a concrete carrier in ETH: 94.1% of multi-asset portfolios hold it, and the median account's return follows ETH's across snapshots (correlation 0.980). Since the market index is split equally between wETH and WBTC, whether accounts beat it depends largely on whether ETH outperforms WBTC, which it did in only 41.7% of snapshots: in those, a median of 60.7% of accounts beat the market, against 26.7% in the others (Appendix D.1).

Taken together, these findings challenge the descriptive adequacy of MPT in the cryptoasset domain. Decentralised markets are natural empirical testbeds for MPT, yet actual allocations bear little resemblance to mean-variance optimal portfolios. Pervasive under-diversification and largely static holdings suggest risk-return optimisation is not a salient feature at the wallet level. The dominance of market-wide movements over allocation choice further places returns in a regime governed by a common market factor [34], in which the diversification benefits MPT promises are structurally limited. Mean-variance optimisation thus appears neither descriptive of observed behaviour nor prescriptively useful in this setting, even if MPT remains a normative benchmark.

Our findings have practical implications. Our empirical analysis suggests that rebalancing toward MPT optima does not reliably improve risk-adjusted performance, while passive market-cap

weighting is observed to offer a more robust default, although on Ethereum it largely amounts to concentrating on ETH. These findings could be of use to whoever wishes to import the mean-variance framework to the blockchain setting (e.g., *retail investors or wealth managers*): doing so wholesale risks projecting an illusion of optimisation onto a return regime in which market-wide movements leave little for diversification gains. For *platform and protocol designers*, the persistence of single-asset and unchanged portfolios among smaller accounts suggests that frictions (e.g., fees, limited attention) shape observed behaviour more than MPT. So, interventions targeting transaction costs or portfolio-construction interfaces may have greater practical impact than tools encouraging frontier-based optimisation. For *regulators and supervisors*, the documented wealth concentration raises concerns about systemic exposure that allocation-focused frameworks do not address.

7.2 Limitations and Future Work

Our analysis is subject to several limitations, each of which points to directions for future work.

First, we cover only the Ethereum blockchain and tokens with minimum market activity [6, 17] and whose balances follow from transfer-like events [31, 41, 50]. Given our focus on portfolios, addresses that have only ever held ETH lie outside our account universe, so the reported single-asset share and degree of under-diversification are lower bounds. ETH staked on the beacon chain and assets held on custodial exchanges are not part of the observed portfolios. Extending the reconstruction to other EVM-compatible blockchains can broaden coverage.

Second, we treat each Ethereum address as an independent portfolio. Linking addresses to common entities remains an open problem in blockchain analytics [11, 33, 51]; off-chain heuristics such as address clustering cannot guarantee full accuracy [40, 49]. Entity-level portfolios may therefore differ in size and composition from the address-level portfolios we observe, and yield a complementary view at the actor level.

Third, we evaluate portfolios against MPT benchmarks rather than underlying decision-making process (e.g., trading motives, herding, disposition effects). While most accounts underperform the market benchmark, characterising the minority of outperformers could reveal whether systematic strategies or informational advantages exist on-chain. The publicly available reconstruction pipeline (and dataset upon publication) provide a foundation for such analyses.

8 Conclusion

We reconstructed complete cryptoasset portfolios over 4562 tokens for 116 M Ethereum accounts from the full chain history (2015–2025) and evaluated them at monthly snapshots from 2020 to 2025 against MPT benchmarks. Both the actual on-chain portfolios and their MPT-optimal counterparts underperform the market benchmark in median realised return. On-chain account portfolios are pervasively under-diversified and largely static, and where they sit close to the constrained efficient frontier, they do so by construction. Realised returns are governed by market-wide movements, driven on Ethereum largely by ETH, rather than by *how* accounts allocate. MPT thus appears neither descriptive nor prescriptively useful in this setting, even as it retains its value as a normative benchmark. The publicly available reconstruction pipeline (and dataset upon publication) provides a foundation for further analysis.

Acknowledgments

Omitted for review.

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A Glossary

Following is a list of the terms, notations, and acronyms used in the paper.

A.1 Glossary

covariance matrix	Matrix Σ whose entry (i, j) is the covariance of returns between assets i and j; the diagonal contains per-asset variances. Estimated empirically over a lookback window and used as input to MVO.
cryptoasset	Native ETH or an ERC-20 token, the two asset families that make up the portfolios studied in this paper.
efficient frontier	The set of portfolios that achieve the highest expected return for a given level of volatility. Throughout the paper, the empirically relevant quantity is the *constrained* efficient frontier: we optimise over each account's already-held assets rather than over all available assets, with native ETH and its wrapped form (WETH) treated as a single asset.
Gini coefficient	A measure of statistical dispersion ranging from 0 (perfect equality) to 1 (maximal inequality), used here to quantify wealth inequality among token holders.
naive allocation	A baseline allocation that requires no estimation of returns or covariances. The $1/N$ (equal-weight) variant assigns weight $1/N$ to each of the N held assets; the market-cap variant weights each asset by its market capitalisation. Used here as benchmarks against MPT-optimal allocations.
Sharpe ratio	The ratio of a portfolio's expected excess return to its volatility, measuring risk-adjusted performance.
token	A digital asset issued on a blockchain platform, typically following a standard such as ERC-20 on Ethereum.

A.2 Acronyms

CA	contract account
CAGR	compound annual growth rate
CAPM	capital asset pricing model
DeFi	decentralised finance
EOA	externally owned account
ERC-20	Ethereum request for comments 20
HHI	Herfindahl-Hirschman index
MaxRet	max return at target volatility
MaxSR	max Sharpe ratio
MinVar	min variance at target return
MPT	modern portfolio theory
MVO	mean-variance optimisation
TVL	total value locked
WBTC	wrapped BTC
wETH	wrapped ETH

B Supplementary Material: Portfolio Characteristics

This appendix provides supplementary analysis for the portfolio characteristics discussed in Section 4. It contains additional detail on the evolution of wealth and on account growth trajectories (Section B.1), the decomposition of top holders by account type (Section B.2), address-level attribution of the wealthiest accounts (Section B.3), standard inequality measures confirming the wealth concentration documented in the main text (Section B.4), effective diversification and the effect of dust positions (Section B.5), and month-to-month trading activity (Section B.6).

B.1 Accounts and Wealth

This subsection details the evolution of total wealth, account growth trajectories, the distribution of wealth across bins and account types, and the concentration of value among top holders, supplementing the descriptive statistics in Section 4.

B.1.1 Evolution of wealth. Figure 9 provides an overview of the total on-chain wealth in the ecosystem, native ETH and ERC-20 tokens combined, with its evolution over time at two levels of aggregation, all accounts and the CA subset, contrasted with Ethereum TVL in DeFi protocols as reported by DefiLlama [14].

Total wealth follows the well-known boom–bust–recovery cycle of the broader crypto market, tracking the TVL trajectory but exceeding it by a median factor of 22, with both series peaking in December 2021 before contracting sharply through 2022 as a sequence of market shocks unfolded.[5] From 2023 onwards the two series decouple, with portfolio wealth exhibiting an additional growth trend beyond TVL, potentially due to native ETH held in plain accounts and the rise of meme coins and other tokens held outside DeFi protocols, and peaking at $1.91 T in October 2025, while TVL never regains its 2021 peak and ends at $67.6 B.

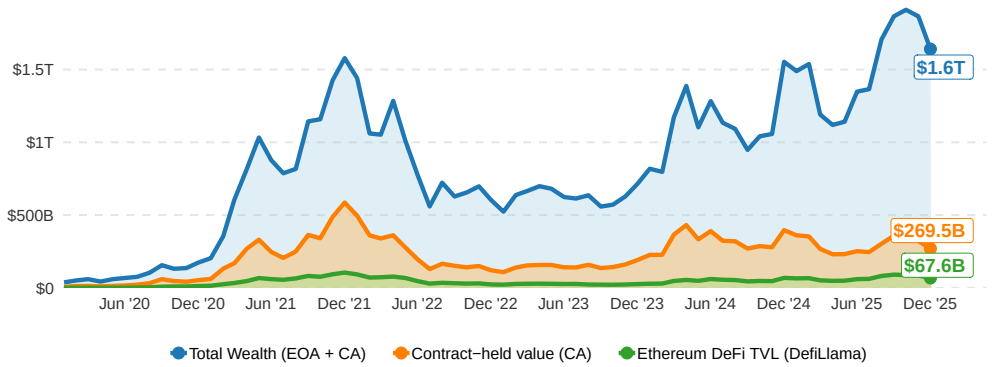


Fig. 9. Total reconstructed on-chain wealth (native ETH and ERC-20 tokens) at monthly snapshots, aggregated over all accounts and over the CA subset, contrasted with Ethereum TVL in DeFi protocols from DefiLlama [14].

The CA share of total wealth (26.26% on average, Section 4.1) is highest during the 2020–2021 DeFi expansion and declines toward end-2025 as wealth shifts to EOA accounts. TVL misses the share held in EOA accounts, but the CA-held value (peaking at $585.9 B in December 2021) is not TVL either; the two are not nested. It is broader in that it covers all deployed contracts, not only DeFi protocols, and is not de-duplicated across receipt and wrapper tokens (aTokens, LP

[5] Terra/LUNA (May 2022), Celsius and Three Arrows Capital (June–July 2022), FTX (November 2022).

tokens, stETH), yet narrower in that it counts only the assets of our universe and excludes ETH staked on the beacon chain, since the deposit contract is not treated as a holder. The former effects dominate: CA-held value exceeds TVL by about 5.6× at the 2021 peak, the gap between the CA and DefiLlama lines in Figure 9. Both series are also subject to double-counting of wrapped or derivative tokens [46].

B.1.2 Account evolution. We analyse the evolution of accounts that have acquired at least one asset of the token selection, and distinguish between EOAs and CAs in the line plot of Figure 10.

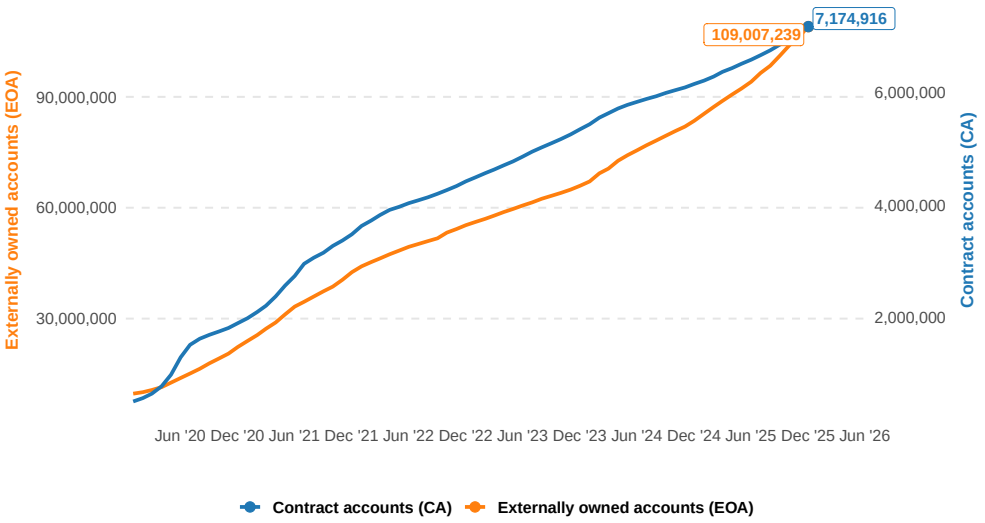


Fig. 10. Number of accounts: the temporal evolution of accounts that have acquired at least one asset.

The two account types exhibit substantially different growth trajectories. By the end of 2025, EOAs reach 109 007 239 addresses against 7 174 916 CAs, so that contract accounts represent only 6.18 % of all accounts that have ever held an asset. Most of these contracts hold no asset at a given snapshot, so among the accounts holding assets at a snapshot, CAs make up on average only 1.33% (Section 4.1). The EOA population grows at a compound annual growth rate (CAGR) of 50.53 % over 2020–2025, adding roughly 16 785 778 addresses per year. CAs grow at a comparable CAGR of 56.93 %, corresponding to approximately 1 128 397 new contracts per year, reflecting the steady deployment of DeFi protocols and token infrastructure.

B.1.3 Wealth distribution of accounts. We classify accounts into wealth bins

$$(0, 1],\ (1, 100],\ (100, 1\text{K}],\ \ldots,\ (100\text{K}, \infty) \quad [\text{USD}],$$

restricting the statistics to active addresses (non-zero wealth); this binning scheme reflects the heavy-tailed nature of on-chain wealth distributions. Table 3 reports the time-averaged share of accounts in each bin separately for CAs and EOAs, while Figures 11 and 12 show the temporal evolution for the low- and high-wealth bins respectively.

Table 3. Wealth bin distribution: time-averaged share of accounts in each wealth bin across the observation window (non-zero USD value accounts only). Shares are percentages.

Type	0–1	1–100	100–1K	1K–10K	10K–100K	>100K
CA	47.52	29.69	11.83	6.68	2.49	1.80
EOA	28.46	53.45	11.59	4.72	1.43	0.35

Across both account types, the majority of accounts consistently hold wealth below \$100, but the within-type composition differs markedly: for CAs the lowest-wealth bin ($\leq$ \$1) holds the largest share (47.52%), whereas for EOAs the \$1–\$100 bin is the largest (53.45%). The share of low-wealth accounts decreases from 2020 to 2021 and rises thereafter, while the share of high-wealth accounts peaks around mid-2021, consistent with the market run-up and contraction described above. The two account types reach their respective minima at different times: the CA minimum around mid-2021 ("DeFi Summer"), whereas the EOA minimum around late 2021 aligns with the broader market peak, together producing the two wealth peaks visible in Figure 9.

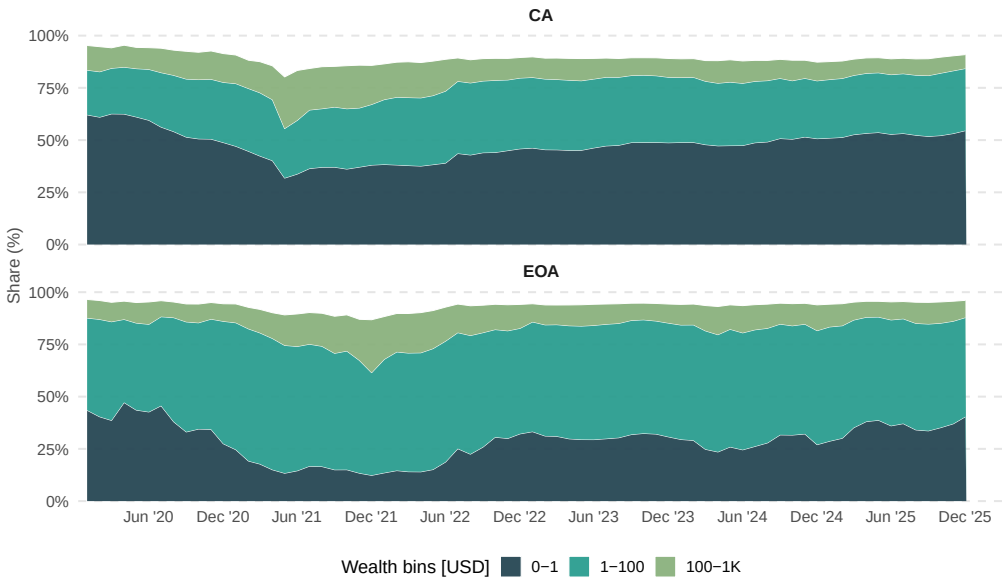


Fig. 11. Distribution of accounts across *low-wealth* bins over time (share of accounts with non-zero wealth). The top subplot shows CA, the bottom subplot EOA.

B.1.4 Wealth of top accounts. Table 4 summarises the yearly concentration statistics for all accounts with portfolio value above \$1, extending the aggregate figures reported in Section 4.1.

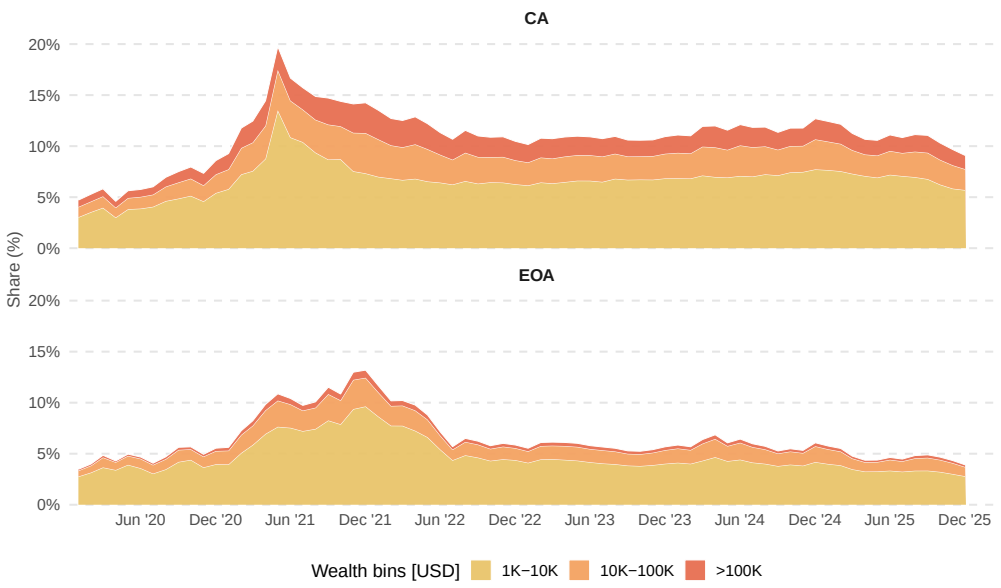


Fig. 12. High-wealth bin distribution over time: share of accounts across high-wealth bins, with CA (top) and EOA (bottom). The peak in high-wealth CA accounts around mid-2021 coincides with DeFi Summer, when protocols experienced rapid TVL growth through liquidity mining.

Table 4. Top-holder concentration by year: yearly averages of top-holder wealth shares for all accounts with portfolio value above $1. Counts are in millions; shares in percent. The Top 1% (min–max) column captures within-year variation across monthly snapshots.

Year	Avg accounts (M)	Top 1% (avg)	Top 1% (min–max)	Top 5% (avg)	Top 10% (avg)
2020	9.015	95.88	94.73–96.57	98.87	99.50
2021	23.120	97.37	96.40–97.72	99.34	99.72
2022	31.743	97.72	97.46–97.91	99.36	99.69
2023	34.747	97.17	97.02–97.28	99.36	99.75
2024	45.291	97.69	97.33–98.05	99.47	99.79
2025	51.320	98.48	98.12–98.79	99.66	99.86

B.2 Account Type Breakdown of Top Holders

To understand whether the extreme wealth concentration documented in Section 4.1 is driven by protocol-controlled accounts or individual holders, we decompose the top 1% and top 0.1% by account type.

Within the top 1%, contract accounts represent 4.08% of accounts by count but hold 26.78% of the top-1% wealth on average. EOAs account for the remaining 95.92% of accounts and 73.22% of value. By count, CAs are thus over-represented, at about three times their 1.33% share of all accounts holding assets. By value, their share matches the ecosystem-wide CA wealth share of 26.26% reported in Section 4.1, as expected because the top 1% holds almost all value.

Narrowing the focus to the top 0.1% sharpens the picture. CAs now constitute 11.90% of addresses, nearly three times their share in the top 1% and about nine times their share of all accounts holding assets, while their value share remains stable at 28.17%. This indicates that the very largest holders include a substantially higher proportion of smart contracts, likely corresponding to DeFi protocol treasuries, exchange hot wallets, and token vesting contracts, while the majority of wealth even at this level is still held by individual EOA addresses.

B.3 Address Attribution of Top-0.1% Holders

To further characterise the entities behind the wealthiest addresses, we match the top 0.1% of accounts at each monthly snapshot against address tags provided by Iknaio [28], queried via the GraphSense REST API [23]. The dataset assigns category labels and semantic concepts to known Ethereum addresses. On average, 20.26% of top-0.1% addresses carry at least one tag (range: 13.48–32.33% across snapshots), while the remaining 79.70% are unlabelled. The limited coverage reflects the pseudonymous nature of Ethereum: the majority of high-value addresses have no publicly attributed identity. Attribution is concentrated in contracts: 75.15% of the contract accounts in the top 0.1% carry a tag against 12.68% of the EOAs, so the unlabelled majority consists almost entirely of EOAs. Among the 20.30% with attribution, three complementary views emerge.

By category. Of the addresses carrying a GraphSense *category* label (7.17% of all top-0.1% block–address pairs), the largest group is *user* (31.15% of categorised), followed by *service* (15.62%), *defi_dex_pair* (15.36%), *defi_token* (10.31%), and *exchange* (8.38%). Together, exchange- and DeFi-related categories account for 40.0% of categorised addresses.

By concept. The broader *concept* taxonomy covers 20.30% of top-0.1% pairs. The dominant concept is *defi* (41.54% of concept-tagged addresses), followed by *exchange* (22.80%), *defi_custody* (20.89%), and *deposit_wallet* (17.23%). This confirms that the top of the wealth distribution is heavily populated by DeFi protocol contracts (liquidity pools, custody vaults, token treasuries) and centralised exchange wallets, consistent with the over-representation of CAs by count reported in Section B.2.

By actor. Where the tags name the operating entity (11.01% of pairs), three actors account for three quarters of the attributions: Gnosis Safe multisig contracts (38.4%), OKX exchange deposit addresses (27.4%), and Uniswap liquidity pools (9.7%); exchanges as a group make up 55.6% of actor-attributed pairs. Exchange deposit addresses also drive the variation of coverage over time: EOA coverage ranges from 7.63% to 26.51% across snapshots, with the high values confined to 2023, when roughly ten thousand OKX deposit addresses temporarily entered the top 0.1%. Only 1.08% of pairs carry an abuse-related tag (blacklists, sanctions, hacks, scams).

B.4 Wealth Inequality and Concentration

Given that the large majority of accounts hold less than $100 on average, we further investigate the wealth distribution to quantify the degree of concentration. We use the Herfindahl-Hirschman Index (HHI) to measure the concentration of wealth among accounts and Gini coefficients to analyse the concentration and inequality of token portfolios. We treat these indices as supplementary rather than primary because address-level inequality measures are debated in pseudonymous blockchain settings, where one entity can control many addresses or one address can serve many entities [11, 51].

Before turning to these indices, we briefly characterise the token-level holder base, which provides context for interpreting concentration measures. Over time, the typical token holder base shrinks: the average number of holders per token decreases from 27 050 (median: 3298) in 2020 to 12 429 (median: 1746) in 2025. This decline is driven by a compositional shift. Figure 13 summarises the annual distribution of tokens grouped by their holder count. The share of tokens with more than

10 000 holders drops from 28.1% to 15.6%, while smaller tokens (100–1 000 holders) grow from 17.7% to 27.5%. Mid-sized tokens (1 000–10 000 holders) remain the dominant group throughout, ranging from 39.8% to 50.4%.

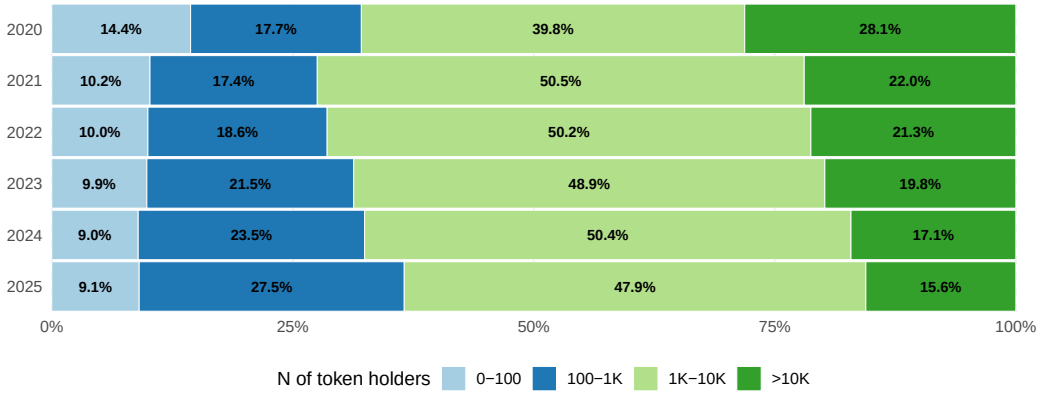


Fig. 13. Token holder distribution: share of ERC-20 tokens grouped by number of holders per token (2020–2025).

To complement the top-holder analysis in Section 4.1, we compute two standard inequality measures for each ERC-20 token at every monthly snapshot and report their temporal evolution. We restrict the analysis to tokens with more than 100 holders at the time of the snapshot to focus on mature assets.

The *Gini coefficient* ranges from 0 (all accounts hold equal wealth) to 1 (one account holds all wealth). The *Herfindahl–Hirschman Index* (HHI) is calculated as $\mathrm{HHI} = \sum_{a=1}^{N} s_a^2$, where s_a is the wealth share of account a and N is the total number of accounts; higher values indicate greater concentration.

Figure 14 displays the average and median of both indices across all qualifying tokens over time. Gini coefficients remain uniformly close to 1 throughout the observation period, with the cross-token average ranging from 0.958 to 0.978 and the median from 0.975 to 0.990. HHI values indicate moderate to high concentration, with the average ranging from 0.218 to 0.295 and the median from 0.138 to 0.220. Both measures are stable over time, confirming that the extreme wealth concentration documented by the top-holder shares in the main text is a persistent structural feature of the ERC-20 token ecosystem.

B.5 Effective Diversification

Weights w_i are the value shares of an account's priced positions, with native ETH and wETH counted as one asset. For every account and snapshot, we compute the largest weight $\max_i w_i$, the HHI of the weights $\sum_i w_i^2$, and the effective number of assets $N_{\mathrm{eff}} = 1/\sum_i w_i^2$, which lies between 1 and N and equals N only for equal weights. As in the main text, all statistics are computed per snapshot and averaged over the 72 monthly snapshots.

Figure 15 shows the distribution of the largest weight and of N_{eff} among multi-asset accounts by wealth category. Concentration is lowest at \$100–\$1,000 and rises towards both ends of the wealth distribution: above \$100,000, most multi-asset accounts keep more than 99% of their value in one asset. Table 5 compares nominal and effective size: accounts with ten or more assets hold 16.0 assets on average but have a median N_{eff} of only 3.17. Across all accounts, 87.8% keep more

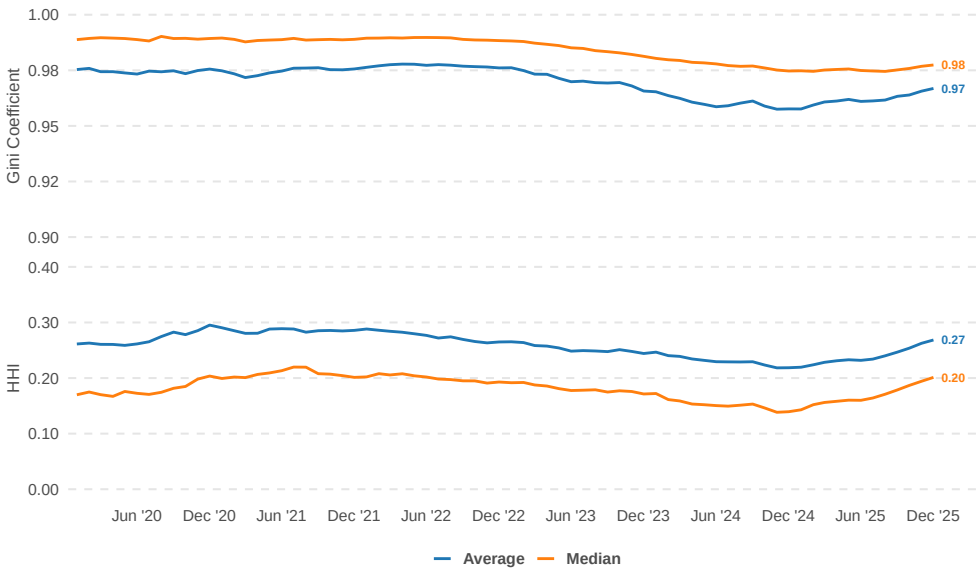


Fig. 14. Inequality measures over time: average and median Gini coefficient (top) and HHI (bottom) across all ERC-20 tokens with more than 100 holders, computed at each monthly snapshot (2020–2025).

than 90% of their value in one asset. The share keeping at least 99% in one asset rose from 72.9% in January 2020 to 83.4% in December 2025, as the share of single-asset accounts grew.

Table 5. Nominal and effective size of multi-asset portfolios, averaged across all monthly snapshots: share of multi-asset accounts, mean number of assets N, median largest weight, median effective number of assets N_{eff}, and share of accounts with at least 99% of their value in one asset.

Assets	Share (%)	Mean N	Median largest weight	Median N_{eff}	≥99% in one asset (%)
2	76.5	2.0	0.94	1.12	32.2
3–4	17.9	3.2	0.82	1.43	16.8
5–9	4.6	6.1	0.64	2.03	5.5
10+	1.0	16.0	0.48	3.17	2.6
All multi-asset	100.0	2.6	0.92	1.18	27.9

Dust positions. The asset count includes every priced position, however small. We therefore repeat the computation ignoring positions worth less than $1. This drops the accounts below $1, which hold no larger position, and keeps every account above $100, so above $100 both computations cover the same accounts (Table 6). Ignoring dust reduces the number of multi-asset portfolios by 37.7% on average (49.0% in December 2025) and the number of portfolios with five or more assets by 39.2% (52.1%). Above $100, more accounts turn out to hold a single asset, while the shares of accounts with at least 99% or more than 90% of their value in one asset do not change: dust inflates the asset count, not the diversification.

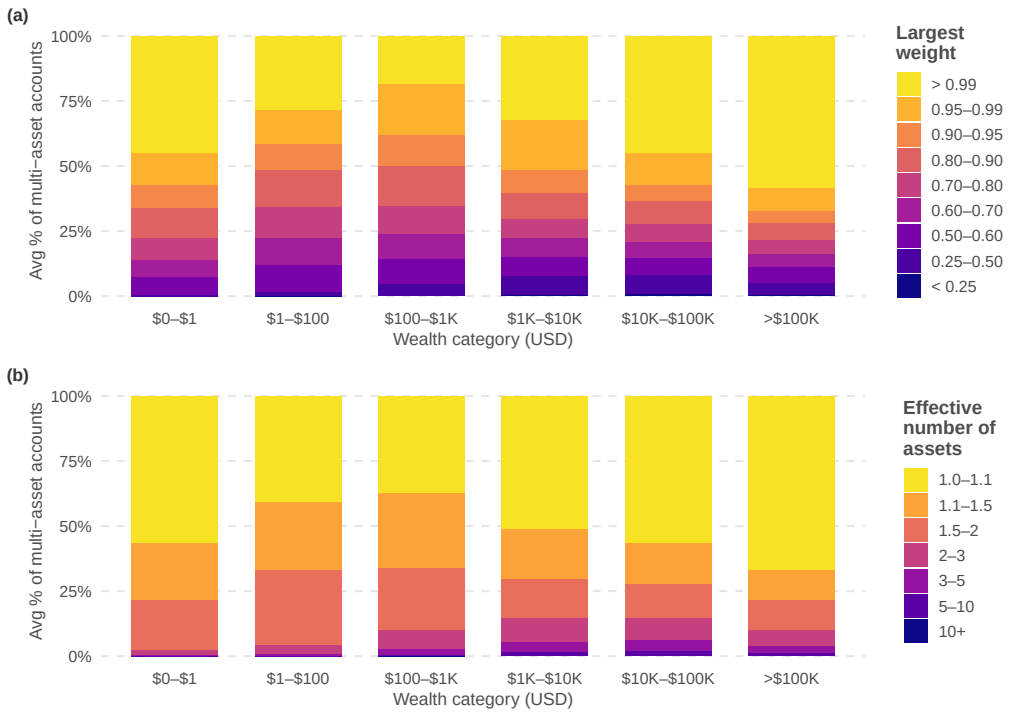


Fig. 15. Concentration of multi-asset portfolios by wealth category, averaged across all monthly snapshots: (a) largest portfolio weight, (b) effective number of assets N_{eff}.

Table 6. Effect of dust positions: portfolio statistics with all priced positions and with positions worth less than $1 ignored, averaged across all monthly snapshots. The upper block covers the accounts above $100, which both computations include (shares of accounts in percent); the lower block gives the number of accounts per snapshot.

	All positions	Positions ≥ $1
Accounts above $100, share of accounts (%)		
One asset	41.0	46.3
Two assets	37.5	37.3
Three to four assets	15.1	11.7
Five or more assets	6.4	4.7
At least 99 % in one asset	56.6	56.6
More than 90 % in one asset	74.0	74.0
$N_{\text{eff}} \geq 2$	6.9	6.8
All accounts, per snapshot		
Multi-asset accounts (million)	11.52	7.17
Accounts with five or more assets (thousand)	700	431

B.6 Trading Activity

For two consecutive snapshots t and t+1, let u_i be an account's units of asset i at t and p_i the asset's price at t+1. The passive weights $\tilde{w}_i = u_i p_i / \sum_j u_j p_j$ are the weights the account would hold at t+1 had it not traded. The turnover $\frac{1}{2}\sum_i |w_i^{(t+1)} - \tilde{w}_i|$ is the share of the portfolio actively reallocated, and the drift $\frac{1}{2}\sum_i |w_i^{(t)} - \tilde{w}_i|$ is the change in weights caused by prices alone. Native ETH and wETH again count as one asset, and an account's holdings are unchanged if no balance differs between the two snapshots. Accounts are grouped by wealth and portfolio size at t; the statistics cover the accounts that hold a positive balance at both snapshots and are averaged over the 71 pairs of consecutive snapshots.

Figure 16 shows the monthly series. Prices alone move the weights of multi-asset portfolios by 2.5% a month on average, and active reallocation amounts to 4.6%; both are small because the portfolios are concentrated. Activity peaked in 2020–2021 and has declined since, also within wealth categories: among multi-asset accounts with \$10,000–\$100,000, the share reallocating more than a quarter of the portfolio fell from 17.6% in 2021 to 8.8% in 2025, and at \$1,000–\$10,000 from 16.5% to 8.1%. Table 7 breaks the activity of multi-asset accounts down by wealth.

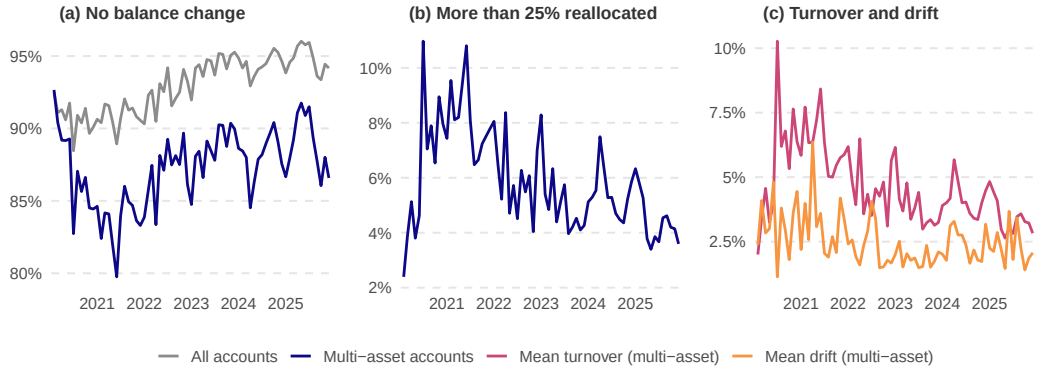


Fig. 16. Month-to-month trading activity: (a) share of the accounts that keep a positive balance and change none of their holdings, for all accounts and for multi-asset accounts; (b) share of multi-asset accounts that reallocate more than 25% of their portfolio; (c) mean turnover and mean drift of multi-asset portfolios. Each point covers the month before the snapshot.

Table 7. Trading activity of multi-asset accounts by wealth category at the first snapshot of each pair, averaged across all pairs of consecutive monthly snapshots: share of accounts without any change in holdings, share reallocating more than 25% of the portfolio, mean turnover, and mean drift (all in percent).

Wealth category	No change	Turnover >25%	Mean turnover	Mean drift
\$0–\$1	97.9	0.9	0.7	2.0
\$1–\$100	93.1	3.4	2.6	2.6
\$100–\$1,000	82.7	8.9	7.0	2.6
\$1,000–\$10,000	73.0	11.6	9.3	2.3
\$10,000–\$100,000	65.0	11.6	9.4	2.0
>\$100,000	59.2	10.7	8.9	1.3
All multi-asset	87.3	5.9	4.6	2.5

Each month, 0.66% of accounts empty their balance, CAs more often than EOAs (1.68% versus 0.64%), and 1.35 million accounts acquire a positive balance, 73.6% of them worth less than $100. These measures are lower bounds on trading, since monthly snapshots miss trades reversed within a month. A transfer is also not always a trade: moving assets to another own address or to an exchange looks like a sale, and CAs such as DeFi liquidity pools change their composition with every swap.

C Supplementary Material: Distance to the Efficient Frontier

This appendix supplements the distance analysis in Section 5. It covers the formal definition of the weight ℓ_1 distance d_s and the expected-return shrinkage applied to return estimates (Section C.1), the calibration and robustness of the risk-free rate r_f (Section C.2), the determinants of d_s and cross-strategy robustness checks (Section C.3), the relationship between d_s and portfolio size with its fitted power-decay model (Section C.4), the comparison with naive allocation benchmarks (Section C.5), and the optimisation volume and temporal evolution of d_s (Section C.6).

C.1 Weight ℓ_1 Distance and Expected-Return Shrinkage

We now provide the formal definition of the ℓ_1 weight distance used throughout the main analysis (Section 5), together with the expected-return shrinkage applied to return estimates in the optimisation pipeline.

Formal definition. Let

$$w_{(0)} = \big(w_{(0),1}, \ldots, w_{(0),N}\big), \qquad w_{(s)} = \big(w_{(s),1}, \ldots, w_{(s),N}\big),$$

with $\sum_{i=1}^{N} w_{(0),i} = \sum_{i=1}^{N} w_{(s),i} = 1$ and $w_i \geq 0$ (long-only). We use ℓ_1 because it has a direct *rebalancing interpretation*: for fully-invested portfolios, $\frac{1}{2}\|w_{(0)} - w_{(s)}\|_1$ equals the minimum one-way turnover (fraction of value traded) required to move from the current to the optimised weights. We therefore define the distance with the $\frac{1}{2}$ factor included (see Definition 3.4).

Using $|u - v| = u + v - 2\min\{u, v\}$ for $u, v \geq 0$ and the fact that both portfolios are fully invested, we obtain

$$d_s = \tfrac{1}{2}\left\|w_{(0)} - w_{(s)}\right\|_1 = 1 - \sum_{i=1}^{N} \min\left\{ w_{(0),i},\ w_{(s),i} \right\}. \tag{3}$$

For long-only portfolios, the distance is bounded by

$$0 \leq d_s \leq 1, \tag{4}$$

where 0 indicates identical allocations and values close to 1 indicate nearly disjoint allocations (mass invested in different assets).

Expected-return shrinkage. Sample mean returns over the $T^{(-)}$-day lookback are noisy in crypto-asset markets, where extreme short-term moves are common. The optimisation pipeline therefore regularises each asset's expected return in two steps. First, outlier days are removed with a trimmed mean: the daily log returns (Definition 3.1) are sorted, the most extreme 5% in each tail are discarded, and μ_i^{trim} is the mean of the remainder. Second, the trimmed mean is shrunk linearly toward the cross-asset average, in the spirit of Bayes–Stein estimation [29]:

$$\mu_i^{\text{shrunk}} = \lambda\, \bar{\mu} \ + \ (1 - \lambda)\, \mu_i^{\text{trim}}, \tag{5}$$

where $\bar{\mu} = \frac{1}{N}\sum_{j=1}^{N} \mu_j^{\text{trim}}$ is the cross-asset mean over the account's asset set and $\lambda \in [0, 1]$ is the shrinkage strength. Sample means are the noisiest input to mean–variance optimisation, and the optimiser amplifies exactly that noise; pulling every estimate halfway toward the grand mean

($\lambda = 0.5$) halves the cross-sectional dispersion the optimiser can chase while preserving the ranking of recent performance. The resulting portfolio expected return is then computed as usual by

$$\mu^{(a,t)} = \sum_{i=1}^{N} w_i^{(a,t)} \mu_i^{\text{shrunk}},$$

which reduces overconfident forecasts while still preserving meaningful cross-sectional differences in recent performance. In practice, this produces smoother optimised weights and more stable allocations across snapshots.

C.2 Calibration and Robustness of the Risk-Free Rate

The MaxSR (Definition 3.3) is known as the *tangency portfolio*, constructed by drawing a line from the risk-free rate r_f tangent to the constrained efficient frontier. It is therefore the only place in our optimisation pipeline where r_f enters; the MinVar and MaxRet optima depend only on the covariance structure and the moments of the actual portfolio. We set $r_f = 0$ throughout — in the MaxSR optimisation and the CAPM alpha evaluation alike — since no genuinely risk-free on-chain asset exists over the study period. This subsection motivates the natural non-zero alternative, $r_f = 5\%$ per annum, on two independent grounds, and shows that the MaxSR results are robust to it.

Reference to the literature. The closest published precedent on portfolio optimisation with crypto assets [45] proxies the risk-free rate with the U.S. 3-month Treasury bill yield. Their convention has carried over to subsequent crypto-portfolio work; in the 2023–2024 window, which contains a large share of our account–snapshot observations, the FRED series `TB3MS` averaged $\approx 5\%$ [5], making 5% the natural non-zero calibration.

Empirical match to on-chain stablecoin yields. Account holders in our sample interact natively with on-chain markets rather than U.S. Treasuries, so we cross-check the T-bill anchor against its closest on-chain analogue: the variable-rate supply APY for USDC and USDT on the major non-custodial money markets active on Ethereum during the sample window. We aggregate the supply APYs reported by DefiLlama's yields API across Aave v2 and v3, Compound v2 and v3, Morpho Blue, and Spark, restricting to pools deployed on Ethereum with TVL above one million USD and APY in $[0, 50]\%$ to exclude inactive or anomalous pools, and average the resulting series within each calendar month from January 2021 onwards. Table 8 reports the unweighted monthly statistics across all months in which lending data are available.

Table 8. On-chain stablecoin supply APY across the major non-custodial money markets on Ethereum (Aave v2/v3, Compound v2/v3, Morpho Blue, Spark), aggregated from DefiLlama's yields API. Unweighted monthly statistics, in %.

Asset	Avg APY	Median monthly APY	Months observed
USDC	5.51	4.77	44
USDT	4.47	3.48	52
USDC and USDT pooled	4.94	3.88	52

The pooled mean of 4.94% essentially matches the T-bill anchor of 5%, so the off-chain and on-chain calibrations point to the same non-zero alternative.

Robustness to the value of r_f. Setting $r_f = 0$ in the MaxSR optimisation is a modelling choice that we now show is empirically harmless. We re-run the full optimisation pipeline with $r_f = 5\%$ on a random sample of 5,000 accounts per monthly snapshot (360,000 accounts across all 72 snapshots) and compare the resulting MaxSR weights and distance metrics, account-by-account, against the production results obtained with $r_f = 0$. The other strategies are unaffected, apart from the few MaxRet instances that fall back to the tangency solution (Section 3.4). Table 9 summarises the per-account differences.

Table 9. Robustness of the MaxSR results to the value of r_f. Per-account quantities computed across 360,000 matched accounts, normalised to the 0–100 scale used throughout the paper.

Quantity	Median	Mean	95th pct (abs)
distance between MaxSR weights ($r_f = 0$ vs. $r_f = 5\%$)	0.0	2.5	5.9
Δd_{MSR} ($w_{(0)}$ vs. MaxSR)	0.0	−0.2	5.5
Δd (MaxSR vs. equal-weight)	0.0	0.0	3.7
Δd (MaxSR vs. market-cap-weight)	0.0	−1.1	5.6

The median account sees no change in any reported quantity, and the mean changes in the three distance metrics stay below two percentage points in absolute value. At the 95th percentile, the MaxSR weights move by 5.9% and the distances by a similar amount, reflecting that the tangency portfolio slides along the same constrained efficient frontier when r_f changes. For the top 1% of accounts, the MaxSR weights move by 80.0% or more: the tangency point jumps to a different corner of the capped frontier, which lifts the mean above the median. Crucially, the change in the headline distance d_{MSR} has a mean of −0.2 pp and a 95th-percentile absolute change of 5.5 pp on a 0–100 scale.

C.3 Determinants and Robustness of Distance

We investigate which observable account characteristics determine distance using the feature set `num_tokens` (the portfolio size N), `log_value_usd`, and `month` on a 10-million-row sample, of which 2 million rows enter the quantile regressions and the Random Forest.

Regression and feature importance. OLS. Linear models explain little variance for MaxSR ($R^2 = 0.062$) and more for MaxRet ($R^2 = 0.205$) and MinVar ($R^2 = 0.330$). In all three, `num_tokens` carries the largest coefficient ($p < 0.001$). An extended specification with polynomial and interaction terms raises R^2 to 0.107, 0.388, and 0.639, respectively. The dominant additional predictor is log(`num_tokens`), confirming the logarithmic relationship between portfolio complexity and distance visible in Figure 7.

Random Forest. A 200-tree model evaluated on a random 20% held-out set achieves $R^2 = 0.505$ (MAE = 9.60 pp) for MaxRet, $R^2 = 0.684$ (MAE = 3.27 pp) for MinVar, and $R^2 = 0.258$ (MAE = 19.65 pp) for MaxSR. Feature importances, shown in Figure 17, diverge sharply. For MinVar and MaxRet, portfolio size dominates with 95.3% and 78.7% of total importance. For MaxSR, importance is spread evenly across `log_value_usd` (35.7%), `month` (33.8%), and `num_tokens` (30.5%). This confirms the qualitative split visible in Figure 7: MinVar and MaxRet distances are largely determined by portfolio size, whereas MaxSR distance additionally depends on wealth and market regime.

Spearman correlations on a 50M-row sample confirm the non-linearity: $\rho = 0.670$ between `num_tokens` and MinVar distance and $\rho = 0.586$ for MaxRet, above the Pearson r of 0.621 and 0.441. For MaxSR the Spearman ρ is 0.214, positive but less dominant.

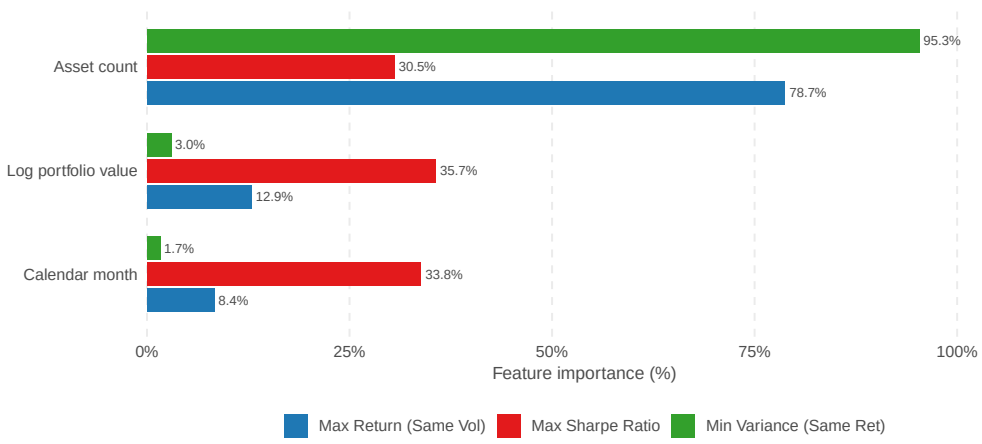


Fig. 17. Random Forest feature importances: impurity-based importances for predicting distance under each optimisation strategy. Portfolio size accounts for 95.3% of importance under MinVar and 78.7% under MaxRet, but only 30.5% under MaxSR.

Quantile regression reveals that at $\tau = 0.25$ and $\tau = 0.5$, only `num_tokens` enters the fitted MaxRet and MinVar quantiles, while the `month` and `log_value_usd` coefficients vanish (low-distance accounts hold exactly two assets regardless of value). At $\tau = 0.75$ the `log_value_usd` coefficient becomes non-zero for MaxRet, indicating that wealth matters primarily in the high-distance tail; for MinVar, the fitted quantiles depend on `num_tokens` alone at all three levels. For MaxSR, all three features enter at every quantile, with wealthier accounts closer to the tangency portfolio. The mass of accounts at zero distance makes the usual standard errors degenerate for MinVar, so we compare coefficients rather than significance levels.

Near-optimal accounts. We examine accounts whose distance falls within 1% of the optimum. Across all three objectives, near-optimal accounts are overwhelmingly simple: 96–97% hold exactly two assets, and cumulative three-or-fewer exceeds 99% in every case. Median portfolio value is modest ($33–$38). The key distinction is the *size* of the near-optimal pool: 641.2M observations (78.4%) are near-optimal under MinVar, 570.2M (69.7%) under MaxRet, and only 146.0M (17.9%) under MaxSR, confirming that minimum-variance and return-maximising optimality are structurally easy to approximate, while Sharpe optimality is difficult even for the simplest allocations.

Per-strategy distance by portfolio size. Figure 18 disaggregates the relationship between portfolio size and distance for each optimisation strategy individually, showing the median (line), mean (dots), and interquartile range (shaded band). The interquartile bands reveal substantial cross-sectional heterogeneity at every portfolio size above two, particularly for MaxSR, where the upper quartile exceeds 80% for every $N \geq 3$, and for MaxRet, whose band spans about 50 percentage points at $N = 5$. The bands narrow as portfolios grow: the largest portfolios are uniformly far from all three optima.

Cross-strategy correlation. The three distance measures are only moderately correlated with one another: $r = 0.594$ between MaxRet and MinVar, $r = 0.366$ between MaxRet and MaxSR, and $r = 0.270$ between MinVar and MaxSR. Being optimally allocated for one objective thus



Fig. 18. Per-strategy distance by portfolio size: median distance (line), mean (dots), and interquartile range (shaded band) as a function of portfolio size N (log-scaled axis), pooled over all 818 M account-month optimisations.

provides limited information about optimality for another, so portfolio "quality" in DeFi is not a unidimensional concept.

Robustness to portfolio size. The preceding results are heavily shaped by the two-asset majority. To test whether the findings generalise to genuinely diversified portfolios, we repeat the full analysis on the subset of account-observations with $N \geq 5$. The near-optimal population essentially vanishes: the share within 1% of the optimum drops from 78.43% to 0.62% for MinVar, from 69.75% to 1.08% for MaxRet, and from 17.86% to 0.20% for MaxSR, confirming that the high near-optimality rates in the full dataset were an artefact of portfolio simplicity rather than genuine optimisation. The distribution also shifts markedly rightward: under MaxSR, 49.2% of $N \geq 5$ accounts fall in the $(80, 100]$% bucket, compared to 12.04% in the full data.

Predictive models lose most of their explanatory power. Random Forest R^2 for MinVar drops from 0.684 to 0.148, and the Spearman ρ between `num_tokens` and MinVar distance falls from 0.670 to 0.339, indicating that the previously dominant portfolio-size effect was largely capturing the gap between two-asset and multi-asset accounts. Feature importances reshuffle accordingly: for MaxSR, `month` becomes the leading predictor (68.4%), suggesting that shifting market regimes make the Sharpe target a moving goalpost for diversified portfolios. The correlation between MaxRet and MaxSR distances also rises from 0.366 to 0.542, implying that among diversified accounts, return-seeking and Sharpe-seeking suboptimality move closer together. Overall, these results reinforce the paper's central finding: observable portfolio characteristics explain suboptimality primarily through the coarse channel of portfolio size, while *which* assets are held (unobserved in our feature set) likely drives distance for any non-trivial allocation.

C.4 Distance by Portfolio Size

Figure 19 disaggregates the binary split from Figure 23 into seven portfolio-size buckets. Across all three strategies, mean distance rises steeply as portfolio size increases from $N = 2$ to $N = 5$ and more slowly thereafter: from $N = 5$ on, each additional asset adds about five percentage points or less. For MinVar, $N = 2$ accounts sit at zero distance, while $N = 5$ accounts already reach more

than half the mean distance of the largest portfolios ($N \geq 21$). The pattern is qualitatively similar for MaxRet and MaxSR, though the absolute distances are higher at every size. This diminishing increase motivates the choice of $N = 5$ as the stratification threshold in the main analysis: below $N = 5$, the small mean distances to MinVar and MaxRet reflect the two-asset portfolios that dominate this group and coincide with these optima by construction, while above $N = 5$, additional assets contribute progressively less further deviation.



Fig. 19. Mean distance over time by granular portfolio-size bucket and optimisation strategy. Labels at the right report the time-series average for each bucket. Distance rises steeply from $N = 2$ to $N = 5$ and more slowly thereafter.

Threshold sensitivity. To verify that the binary split is robust to the choice of cutoff, we repeat the paired Wilcoxon signed-rank test from Section 5.1 for thresholds $N \in \{3, 4, 5, 6, 7\}$. Table 10 reports the results. All fifteen tests reject the null at $p < 1.7 \times 10^{-13}$ (the minimum attainable for 72 paired blocks under the normal approximation), confirming that the small-versus-large contrast is not an artefact of the specific cutoff. At $N = 5$, the threshold used in the main text, Cohen's d is 9.39 for MaxRet, 12.57 for MinVar, and 3.75 for MaxSR, the first two reflecting the near-zero distances that small portfolios attain under MaxRet and MinVar. For MinVar and MaxSR, Cohen's d increases monotonically with N because raising the threshold purifies the "below" group (dominated by two-asset accounts that are structurally close to optimal), thereby widening the gap. The marginal gain in d is largest between $N = 3$ and $N = 5$ and diminishes thereafter, consistent with the flattening visible in Figure 19. For MaxRet, d stays roughly constant across thresholds, because the month-to-month variation of the gap grows roughly in proportion to the gap itself.

Power-decay fit. We model the relationship between portfolio size and mean distance using a three-parameter power-decay specification $\hat{d}_s(n) = \delta_s^{\infty}\,(1 - \psi_s \cdot n^{-\gamma_s})$, estimated by weighted non-linear least squares (trust-region reflective) on 49 portfolio-size bins ($n \in [2, 50]$, minimum 30 observations per bin), with the constraint $\delta_s^{\infty} \leq 100\%$. Each bin's residual is weighted by $\sqrt{N_n}$, where N_n is the number of account-observations at that portfolio size, so that well-populated bins (e.g., n=2 with 621M observations) dominate the fit over sparse tail bins. The model is monotonically increasing and bounded: it equals $\delta_s^{\infty}(1 - \psi_s \cdot 2^{-\gamma_s})$ at the minimum portfolio size and asymptotes

Table 10. Paired Wilcoxon signed-rank tests for the distance gap between portfolios with fewer than N assets and those with N or more, at varying thresholds. All p-values equal 1.66×10^{-13} (Wilcoxon floor for 72 blocks).

Strategy	N	Mean<N (%)	Mean≥N (%)	Δ (pp)	Cohen's d
MaxRet	3	4.3	38.7	34.4	9.17
	4	8.3	49.2	40.8	9.71
	5	9.8	55.4	45.6	9.39
	6	10.6	59.8	49.1	9.60
	7	11.1	63.0	51.9	9.85
MinVar	3	0.0	30.8	30.8	9.75
	4	3.4	42.3	39.0	11.95
	5	4.8	48.7	43.9	12.57
	6	5.5	53.1	47.5	13.13
	7	6.0	56.3	50.3	13.62
MaxSR	3	37.3	56.8	19.6	2.72
	4	39.4	64.5	25.1	3.50
	5	40.3	68.7	28.4	3.75
	6	40.8	71.6	30.7	3.94
	7	41.1	73.7	32.6	4.12

to δ_s^∞ as $n \to \infty$. The asymptote δ_s^∞ need not equal 100%, and none of the three fits reaches that bound.

MinVar and MaxSR plateau at $\delta^\infty = 81.66\%$ and $\delta^\infty = 90.88\%$ respectively, indicating that even infinitely diversified wallets retain some structural overlap with the minimum-variance and Sharpe-optimal allocations. For MaxRet, the fitted $\delta^\infty = 80.60\%$ is set by the steep rise at small n and lies below the empirical means of the largest portfolios, so the fit does not identify a MaxRet ceiling. The exponent γ_s controls the saturation speed. For MaxRet, $\gamma = 1.0033$ is the largest, matching the steep climb from near-zero distance at $n = 2$ but not the continued rise of the sparse tail. MaxSR ($\gamma = 0.8401$) and MinVar ($\gamma = 0.8030$) saturate more gradually, and both still rise noticeably beyond $n \approx 20$.

Table 11 reports the goodness of fit: the observation-weighted R^2 and the unweighted mean absolute error (MAE) across the 49 size bins. Figure 20 overlays the fitted curves on the empirical means. The fits track the data within about one percentage point up to $n = 10$, which covers more than 99% of account-observations, and understate the empirical means in the sparse tail, most for MaxRet, whose tail rises above its fitted asymptote.

Table 11. Goodness of fit of the power-decay distance model (Equation (2)), by optimisation strategy.

Strategy s	R^2	MAE (pp)
MaxRet	0.9993	4.08
MinVar	0.9999	1.69
MaxSR	0.9999	1.35

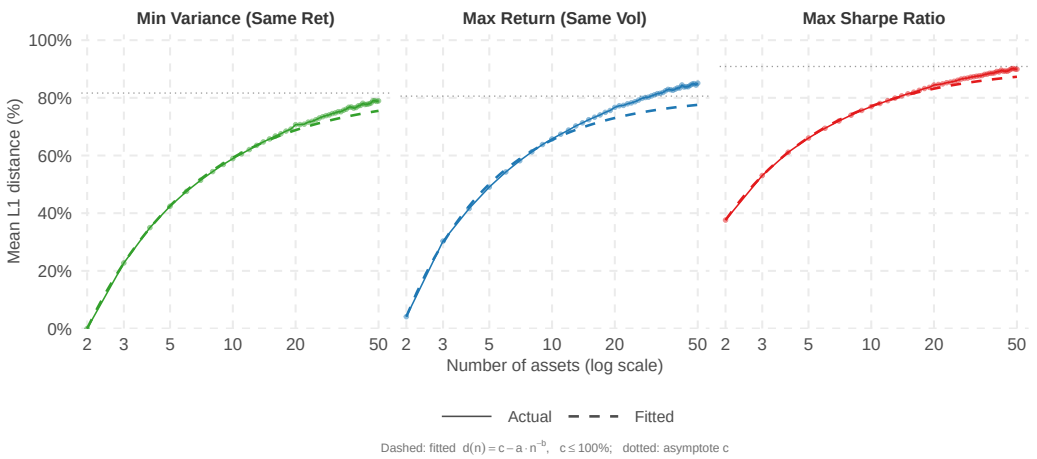


Fig. 20. Fitted versus empirical distance: empirical mean distance (solid) versus fitted power-decay model $\hat{d}_s(n) = \delta_s^{\infty}\,(1 - \psi_s \cdot n^{-\gamma_s})$ (dashed) for each strategy. Dotted lines mark the asymptote δ_s^{∞} where $\delta_s^{\infty} < 100\%$.

C.5 Naive Benchmark Comparison

Table 12 reports the distance between the actual allocation $w_{(0)}$ and two naive benchmarks. Both naive strategies are substantially farther from $w_{(0)}$ than the closest MPT strategy (MinVar), confirming that on-chain accounts do not simply replicate either equal-weight or market-cap-weight allocations. The mean–median relationship differs across benchmarks: for equal-weight the median exceeds the mean, indicating a left-skewed distribution with a concentration of accounts at the 40.0% mark (the distance from a two-asset portfolio projected onto the 90% weight cap to the equal split), whereas for market-cap-weight the mean exceeds the median, reflecting a right tail of accounts that deviate substantially from cap-weighting.

Table 12. Distance to naive benchmarks: distance between the actual allocation $w_{(0)}$ and naive allocation benchmarks, expressed as percentages. The closest MPT strategy (MinVar) is included for reference.

Benchmark	Mean d_s (%)	Median d_s (%)
Equal-weight ($1/N$)	36.0	40.0
Market-cap-weight	41.9	33.5
MinVar (reference)	*7.11*	*0.00*

Table 13 and Figure 21 decompose the structural effect of each MPT strategy on distance to the naive benchmarks. Three patterns stand out. First, MinVar is the most conservative optimiser: it leaves the majority of accounts unchanged against both equal-weight (78.2%) and cap-weight (78.1%), and when it does move accounts, it predominantly moves them closer to equal-weight (closer-to-farther ratio ≈ 2.8:1). Second, MaxRet shows a split personality: it moves accounts closer to equal-weight (mean $\Delta d = -1.02$ pp) but farther from cap-weight (mean $\Delta d = +1.20$ pp), indicating that return-maximising portfolios tend to diversify away from concentrated large-cap holdings. Third, MaxSR moves the most accounts away from both benchmarks, 40.8% from equal-weight and 43.1% from cap-weight, against 25.0% and 39.5% that move closer, reflecting the highly specific asset weightings required to maximise the Sharpe ratio.

Table 13. Directional effect of MPT optimisation on naive-benchmark distance: $\Delta d < 0$ indicates the optimised portfolio moved closer; $\Delta d > 0$ farther. Mean Δd in percentage points of the 0–100 distance scale; accounts with $|\Delta d| \leq 0.05$ pp are classified as unchanged.

Strategy	Naive Benchmark	Mean Δd (pp)	Closer (%)	Farther (%)	Unchanged (%)
MaxRet	Equal-weight	−1.02	17.0	12.0	71.0
	MCap-weight	+1.20	14.9	15.8	69.3
MinVar	Equal-weight	−2.06	16.1	5.7	78.2
	MCap-weight	+0.44	10.2	11.8	78.1
MaxSR	Equal-weight	+3.88	25.0	40.8	34.2
	MCap-weight	+3.18	39.5	43.1	17.3

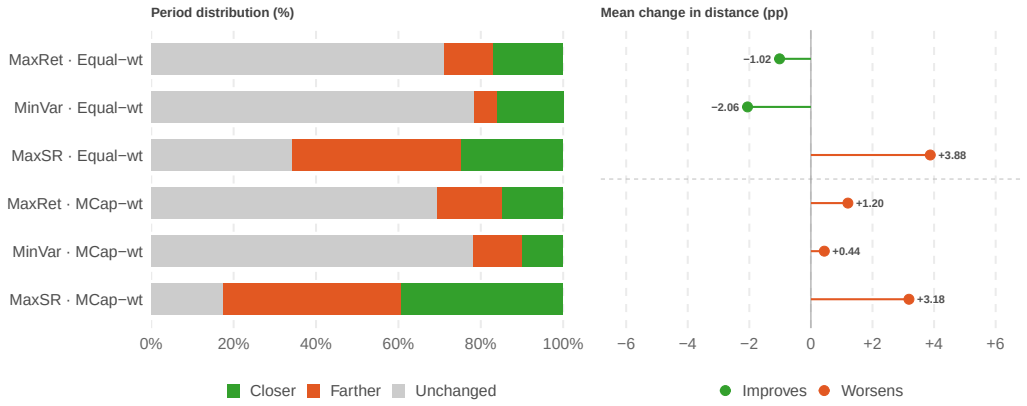


Fig. 21. MPT optimisation versus naive benchmarks: left, proportion of accounts moved closer, farther, or left unchanged; right, mean Δd in percentage points (negative = toward the benchmark; positive = away).

C.6 Optimisation Volume and Temporal Evolution

This subsection reports the optimisation volume and temporal distance statistics underlying the main analysis in Section 5. The 817,525,786 account-month optimisations cover between 3,917,721 and 21,388,124 accounts per monthly snapshot (median 11,037,006).

Aggregate temporal evolution of distance. Figure 22 reports the aggregate mean and median distance, together with ±1 standard-deviation bands, for each optimisation strategy over the full sample period. All three strategies exhibit a mild upward drift in mean distance over time, both overall and among large portfolios. The persistent and substantial gap between mean and median under MinVar and MaxRet reflects the distributional asymmetry of their distances: a large mass of accounts with near-zero distance (pooled ≤ 1% share: 78.43% and 69.75%) coexists with a thin but heavy right tail, pulling the time-averaged means of 7.11% and 12.24% well above medians of zero. Under MaxSR, by contrast, the median (43.94%) lies slightly above the mean (41.86%), as most of its mass sits at large distances. The broad standard-deviation bands across all strategies confirm the high cross-sectional heterogeneity that motivates the portfolio-size stratification of Section 5.1, shown in Figure 23: once accounts are split by portfolio size, the MinVar mean ranges

from 48.7% ($N \geq 5$) to 4.8% ($N < 5$), a spread of more than 40 percentage points that the aggregate view entirely obscures.



Fig. 22. Aggregate distance over time: monthly mean and median with ±1 standard-deviation bands (clipped to $[0, 100]$%) for each strategy, for all accounts and by portfolio size (N = 817 525 786, 72 monthly snapshots).

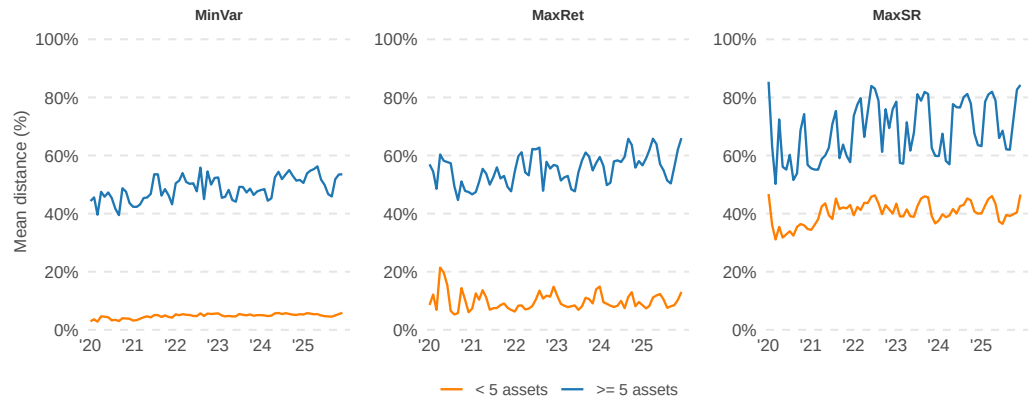


Fig. 23. Mean distance over time by portfolio size. Trajectories for portfolios holding fewer than five assets versus five or more, one panel per optimisation strategy.

D Supplementary Material: Realised Portfolio Performance

This appendix supplements the realised performance analysis in Section 6. It covers the frequency of outperforming the market and the cross-sectional magnitude of beat-market margins (Section D.1), the association between frontier distance and risk-adjusted return (Section D.2), the Random Forest and SHAP analysis of return and alpha predictability (Section D.3), and the out-of-sample validation of the month effect (Section D.4).

D.1 Frequency of Outperforming the Market

Table 2 in Section 6.1 summarises realised returns and alpha by strategy. As a complementary view, we report the within-snapshot share of accounts that beat the wETH and WBTC market index and summarise these per-snapshot percentages across the 72 monthly snapshots. For each strategy and each monthly snapshot we compute (i) the share of accounts whose realised 20-day return exceeds the market and (ii) the share whose CAPM alpha is strictly positive. Both numerator and denominator are restricted to accounts where the relevant return and beta values are present, so each strategy is evaluated on its own valid subpopulation. The baseline filter from Section 6.1 is applied throughout.

Table 14 reports the resulting distributions. Three patterns stand out. First, cross-strategy variation is narrow: across the actual, MPT, and equal-weight allocations, the per-snapshot median ranges from 38.1% to 42.6% by raw return and from 34.3% to 45.7% by alpha, a spread of 4.6 and 11.4 percentage points respectively. Market-cap-weighting is the exception: as cap-weighting concentrates nearly every account on ETH, almost all or almost none of its portfolios beat the market or earn a positive alpha in a given month (fewer than 10% beat the market in 35 snapshots and more than 90% in 27; for positive alpha, 37 and 32 snapshots), so its median across snapshots says little about a typical month. Second, cross-snapshot variation is an order of magnitude larger: the baseline rate alone spans 3.32% to 90.44% across snapshots, and pairwise Pearson correlations of the per-snapshot series across strategies have a median of 0.831 (lowest 0.436, equal- versus cap-weighting), indicating that strategies largely move together over the calendar. Third, in 55.6% of snapshots fewer than half of baseline accounts beat the market index. This pattern follows ETH, which 94.1% of multi-asset portfolios hold: ETH beat the wETH–WBTC blend in only 41.7% of snapshots, and in those snapshots a median of 60.7% of baseline accounts beat the market, against 26.7% in the others. Together these patterns reinforce the conclusion of Section 6.1: the choice of allocation strategy has at most a marginal effect on whether an account beats the market in any given month, while the prevailing market regime explains the bulk of the variation.

Measured against ETH itself rather than the market index, accounts fare similarly: in the median snapshot, 38.96% of baseline accounts beat ETH's 20-day return, and the median excess return over ETH is −0.48 pp, against −0.88 pp over the market index. The cap-weighted portfolio stays much closer to ETH than the actual allocation (Q_{01} to Q_{99} of the excess return over ETH: −15.82% to 9.82%, against −30.00% to 38.28%), and its median return follows ETH's across snapshots (correlation 0.987); so does the median return of the actual allocation (0.980), even though individual accounts deviate from ETH far more.

The frequency view summarised in Table 14 captures only whether each account beat the market in a given month, not by how much. As a complementary magnitude view, for each monthly snapshot we compute the within-snapshot percentiles of the beat-market margin across accounts and report the median of each percentile across the 72 monthly snapshots. Two distributional metrics are reported in parallel: the raw excess return $r_s - r_{(m)}$ and the cross-sectional CAPM alpha $r_s - \beta_s r_{(m)}$. The per-strategy distributions are visualised in Figures 24 and 25, and selected percentiles are reported in Table 15.

Table 14. Per-snapshot frequency of beating the wETH–WBTC market index, summarised across the 72 monthly snapshots. Each cell is the median across snapshots of the within-snapshot share of accounts whose realised return exceeds the market (left) or whose CAPM α is strictly positive (right).

Strategy	Beats market by return (%)	Positive CAPM α (%)
Baseline	40.64	34.34
MaxRet	38.05	45.04
MinVar	39.91	43.29
MaxSR	42.62	45.73
Equal-weight	42.00	41.14
MCap-weight	33.54	8.21

Three patterns are immediately visible. First, every strategy except market-cap-weighting produces a right-skewed cross-sectional distribution: the upper tail extends further from zero than the lower tail. For the baseline portfolio the top one percent of accounts beat the market by 38.62% in a typical month while the bottom one percent return 29.76% below it, an asymmetry of 8.87 percentage points ($Q_{99} + Q_{01}$, with positive values indicating right skew). The asymmetry reproduces qualitatively for the three MPT strategies and equal-weighting and reflects the well-documented right skew of token-level returns on Ethereum, which dominates the cross section even after aggregation to the account level. Second, the centre of mass lies below zero for every strategy: median excess return ranges from −0.72% for MCap-weight to −1.73% for MaxRet, restating the conclusion of Table 14 as a location parameter rather than a binary frequency. Third, the cross-strategy contrasts that the figure makes most visible all live in the tails rather than at the median. MaxSR has the deepest lower tail (Q_{01} = −30.70%), reflecting the classical Markowitz problem of concentrated bets on noisy historical covariances. Equal-weight has the shallowest lower tail of the remaining strategies (Q_{01} = −24.37%) without giving up upper-tail upside (Q_{99} = 40.70%). MCap-weight compresses the whole distribution toward zero (Q_{01} = −15.16%, Q_{99} = 11.69%, skew −3.46 pp), consistent with its alignment with ETH, which makes up half of the market index.

The CAPM α distribution shown in Figure 25 mirrors the excess-return shape closely, with slightly more pronounced tails after the β adjustment (baseline Q_{01} = −31.43% versus −29.76% on raw excess return, and Q_{99} = 38.80% versus 38.62%), indicating that accounts in both tails carry larger absolute β exposure to market moves. The cross-strategy ranking of the tails is unchanged; at the median, the β adjustment moves the three MPT strategies ahead of the actual allocation.

Table 15. Distribution of the per-account beat-market margin, summarised across the 72 monthly snapshots. For each monthly snapshot we compute the listed within-snapshot percentiles across accounts and report the median of each percentile across snapshots. The *Skew* column is $Q_{99} + Q_{01}$. Positive values indicate right skew, and values are reported in percentage points.

	Excess return $r_s - r_{(m)}$ (%)				CAPM α $r_s - \beta_s r_{(m)}$ (%)			
Strategy	Q_{01}	Q_{50}	Q_{99}	Skew	Q_{01}	Q_{50}	Q_{99}	Skew
Baseline	−29.76	−0.88	38.62	8.87	−31.43	−0.64	38.80	7.38
MaxRet	−29.40	−1.73	41.62	12.22	−30.07	−0.36	41.16	11.09
MinVar	−27.81	−1.03	39.71	11.90	−28.52	−0.20	39.37	10.85
MaxSR	−30.70	−1.35	44.08	13.38	−31.63	−0.36	48.40	16.77
Equal-weight	−24.37	−1.48	40.70	16.33	−24.79	−0.65	40.66	15.88
MCap-weight	−15.16	−0.72	11.69	−3.46	−13.67	−0.27	7.03	−6.64

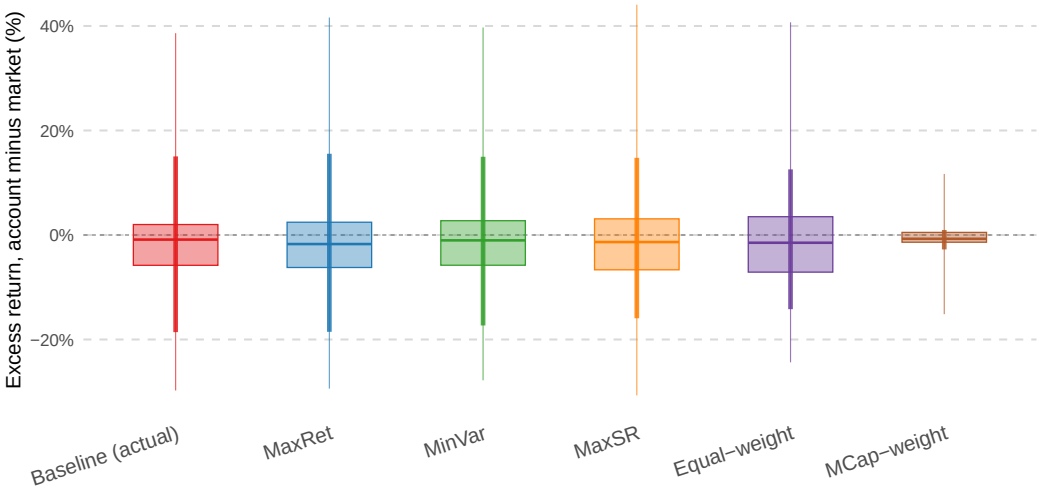


Fig. 24. Cross-sectional distribution of the per-account excess return $r_s - r_{(m)}$, by strategy. Each box summarises the within-snapshot percentiles across accounts (filled box Q_{25} to Q_{75}, inner line Q_{50}, thick whisker Q_{05} to Q_{95}, thin whisker Q_{01} to Q_{99}), aggregated as the median of each percentile across the 72 monthly snapshots. The dashed line marks the market.

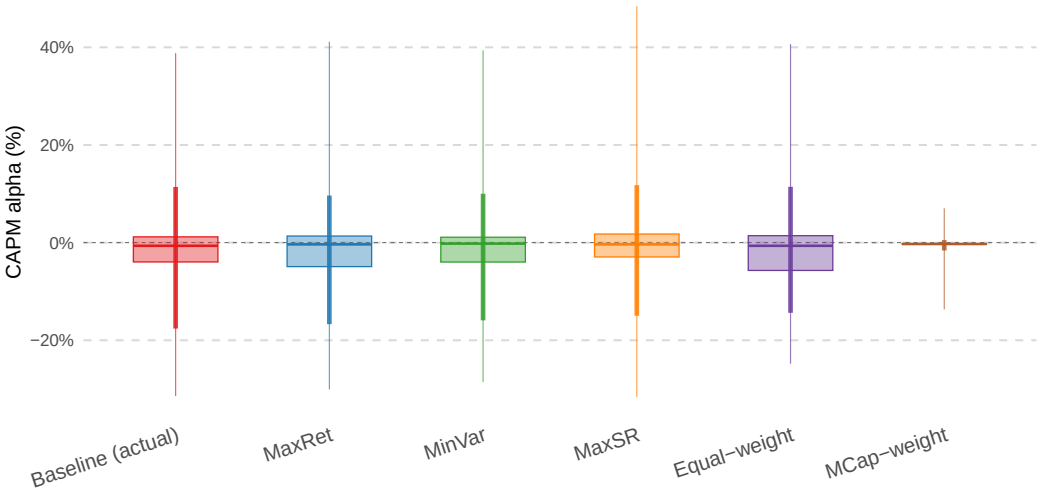


Fig. 25. Distribution of the per-account CAPM α $r_s - \beta_s\, r_{(m)}$ by strategy, with quantiles aggregated as the median across all 72 monthly snapshots. Boxes are constructed as in Figure 24, with the dashed line marking zero alpha.

As a complementary temporal view, Figure 26 contrasts the between-strategy return differences with the magnitude of market-wide movements across snapshots, underscoring that allocation choice is a second-order effect relative to the market cycle.

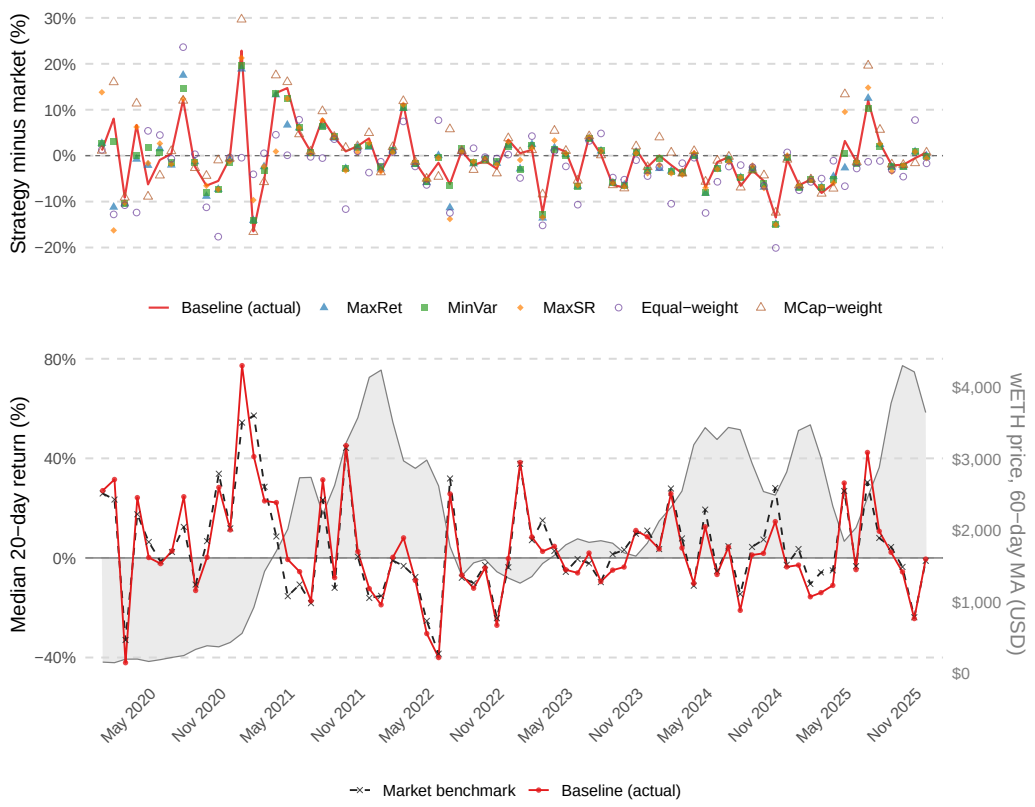


Fig. 26. Returns relative to the market. Top: median $T^{(+)}$-day return minus the market benchmark per snapshot, showing that allocation choice has little effect relative to market-wide movements. Bottom: median baseline return and market benchmark overlaid on the wETH 60-day moving average price. The spread between the six allocations has a median of 4.0 pp per snapshot, while the market return swings over 95.7 pp.

D.2 Frontier Distance and Risk-Adjusted Return

In Section 5 we showed that portfolio size is the dominant determinant of ℓ_1-distance from the efficient frontier. If MPT optimality matters in practice, we would expect a negative association between distance (d) and risk-adjusted returns (α). We find at most a negligible one: Spearman rank correlations between ℓ_1-distance and baseline α are negative but close to zero ($\rho = -0.060$ for MinVar and -0.058 for MaxSR), so distance accounts for less than half a percent of the rank variation in α, and the three distances add no out-of-sample information on α within a month (Section D.4). The raw return is higher for zero-distance accounts (-0.45%) than for the 80–100 % bucket (-5.15%), but this gap is structural rather than a payoff to optimisation: as Sections 4.2 and 5 showed, near-optimal accounts are almost exclusively two-asset portfolios that mechanically land close to the frontier.

D.3 Random Forest and SHAP Details

To quantify the relative importance of account characteristics for predicting performance, we train Random Forest regressors (300 trees, depth 18, at least 100 accounts per leaf) on a sample of

50,000 accounts per monthly snapshot (3.6 M account-snapshots, targets winsorised at the 5th and 95th percentiles within each snapshot) on two target variables—realised $T^{(+)}$-day return (%) and cross-sectional α—for each of the six strategies. We consider two feature sets: a *baseline* model with three features (snapshot month, log portfolio value V, portfolio size N) and an *extended* model of six features that adds portfolio β, a binary indicator for holding ETH (native, wrapped, or liquid-staked), and a binary BTC-wrapper indicator.

Feature importance. Table 16 reports the permutation-based feature importances (the loss of held-out R^2 when a feature is shuffled, as a share of the total) and R^2 for both models on the baseline strategy. The pattern is consistent across all six strategies (full results in the accompanying data release): month dominates, β is a meaningful but secondary predictor, and asset composition is negligible.

Table 16. Random Forest feature importances: permutation-based importances and out-of-sample R^2 for predicting baseline returns and α. The 6-feature model adds portfolio β and the ETH and BTC-wrapper indicators.

	Return (%)		α	
Feature	3-feat	6-feat	3-feat	6-feat
Snapshot month	79%	64%	58%	44%
Portfolio β	—	20%	—	30%
Log portfolio value	21%	15%	41%	25%
Portfolio size N	1%	<1%	1%	<1%
Holds ETH	—	1%	—	1%
Holds BTC wrapper	—	<1%	—	<1%
R^2 (test)	0.73	0.88	0.57	0.79

SHAP analysis. We compute SHAP [36] values via TreeExplainer [35] on a 10 K subsample to assess how each feature shifts individual predictions. Figure 27 shows the SHAP summary for baseline returns. Snapshot month dominates, shifting predicted returns by ±15.0 pp on average. The portfolio β contributes ±2.9 pp, portfolio value ±1.1 pp, and all remaining features at most ±0.2 pp.

The SHAP dependence plot for snapshot month (Figure 28) shows predicted returns swinging by tens of percentage points across the sample period. The per-month median (red line) tracks the broader crypto market cycle. The model is not learning which accounts allocate well, rather, which months were good or bad to be in the market.

3-feature vs. 6-feature comparison. Adding β and asset-composition indicators improves R^2 by +0.14–+0.17 for returns and +0.21–+0.28 for α across all strategies but market-cap-weighting (Table 17), whose outcomes the month alone predicts almost perfectly, because the cap-weighted portfolio is close to ETH for nearly every account. The improvement is driven almost entirely by β: the two binary asset indicators contribute at most 3% of importance in any model. This confirms that β carries genuine predictive content beyond timing and wealth, but the snapshot month remains the dominant factor in both specifications.

D.4 Out-of-Sample Validation of the Month Effect

The random 80/20 split of Section D.3 places every snapshot month in both the training and the test data, so a high month importance shows that the models have learned each month's average

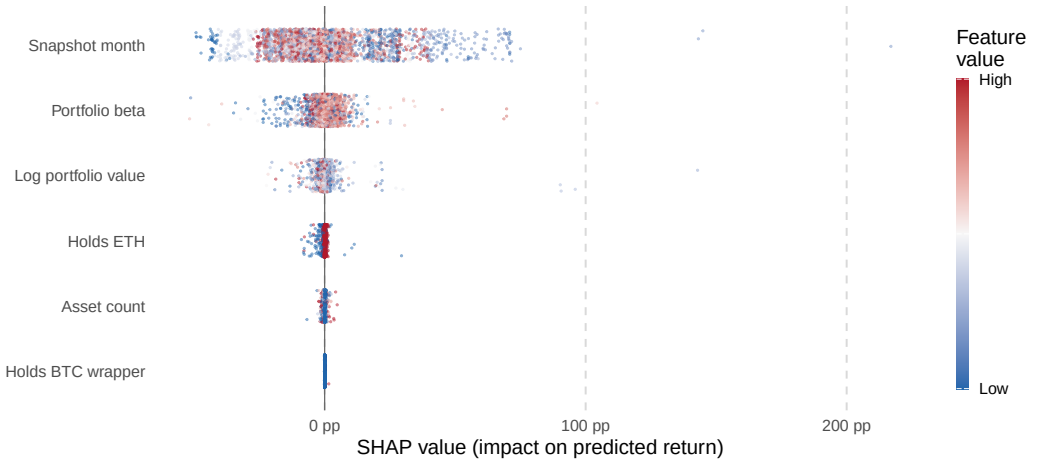


Fig. 27. SHAP summary for baseline returns. Each dot represents one account-snapshot, the x-axis shows feature contribution to predicted return (pp), and colour encodes feature value (blue = low, red = high). Snapshot month dominates (±15.0 pp on average).

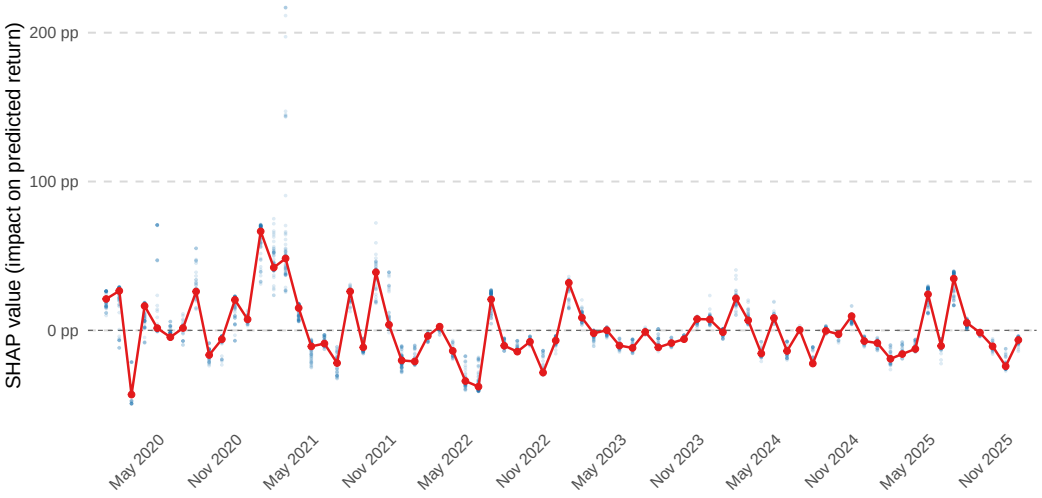


Fig. 28. SHAP dependence for snapshot month. Each dot is one account-snapshot, the x-axis is the calendar month of the snapshot, and the y-axis the SHAP value (pp). The red line marks the per-month median.

outcome, not that this outcome could have been anticipated. We therefore split the variance of the baseline outcomes into a between-month and a within-month part, and re-validate the models with whole months held out of training (Table 18).

Variance decomposition. Differences between monthly means account for 55.6% of the variance of 20-day returns across all account-snapshots (50.7% with every month weighted equally), but only for 19.4% (18.2%) of the variance of α. Returns thus share a large component common to all

Table 17. Model comparison across strategies: out-of-sample R^2 for 3-feature versus 6-feature Random Forest models (random 80/20 split).

	Return (%)			α		
Strategy	3-feat	6-feat	Δ	3-feat	6-feat	Δ
Baseline	0.73	0.88	+0.15	0.57	0.79	+0.22
MaxRet	0.71	0.85	+0.14	0.53	0.74	+0.21
MinVar	0.73	0.88	+0.15	0.57	0.78	+0.21
MaxSR	0.71	0.87	+0.16	0.48	0.75	+0.27
Equal-weight	0.70	0.88	+0.17	0.50	0.79	+0.28
MCap-weight	0.99	1.00	+0.01	0.96	0.99	+0.03

accounts in a month, the market-wide movement, whereas α, which removes the market exposure, is mostly specific to the account.

Month-grouped validation. We split the 72 snapshots into six folds of whole months and test each model on the months it has not seen. The six-feature model of Section D.3 then reaches $R^2 = -0.44$ for returns and -0.29 for α (variant b), worse than a constant prediction. Replacing the month by nine ex-ante market-state features (variant c)—the Crypto Fear & Greed Index[6] and its extreme-fear and extreme-greed flags, the 30- and 90-day momentum and the 30-day volatility of the market index, its 365-day drawdown, the growth of the account base, and the previous snapshot's median α—gives $R^2 = -0.04$ and -0.03, with leaves of at least six months of accounts so that the forest cannot memorise single months. In a forward-chaining test that trains on all earlier years, the out-of-sample R^2 of this model per test year (2022–2025) ranges from -0.66 to 0.35 for returns and from -0.39 to 0.12 for α, so no stable relation emerges. Finally, removing each month's mean from the target and adding the three frontier distances to the account features (variant d) leaves the within-month variation unexplained ($R^2 = -0.09$ and -0.06).

Table 18. Random Forest variants for baseline returns and α. Account features: log portfolio value, portfolio size N, portfolio β, and the ETH and BTC-wrapper indicators. Variants b–d are tested on whole months held out of training (six month-grouped folds).

Variant	Features	Validation	R^2 return	R^2 α
(a)	month + account features	random 80/20	0.88	0.79
(b)	month + account features	month-grouped	−0.44	−0.29
(c)	market state + account features	month-grouped	−0.04	−0.03
(d)	account features + distances (demeaned target)	month-grouped	−0.09	−0.06

Together, these tests show that the month effect reflects market-wide movements that observable market conditions do not predict, and that within a month the allocation, frontier distance included, carries no predictive information about realised returns.

E Reconstruction and Computational Details

This appendix details the native ETH reconstruction and describes the computational infrastructure and software stack used to produce the results presented in this paper, supplementing the

[6] https://alternative.me/crypto/fear-and-greed-index/

methodological overview in Section 3.3. It covers the native ETH event taxonomy and ledger ordering (Section E.1), the execution environment (Section E.2), the four-stage computational pipeline (Section E.3), and the software stack (Section E.4).

E.1 Native ETH Reconstruction

This subsection details the native ETH route of Section 3.3 (Figure 29): the source tables, the complete taxonomy of balance-changing events, and the intra-block ordering of the ledger.

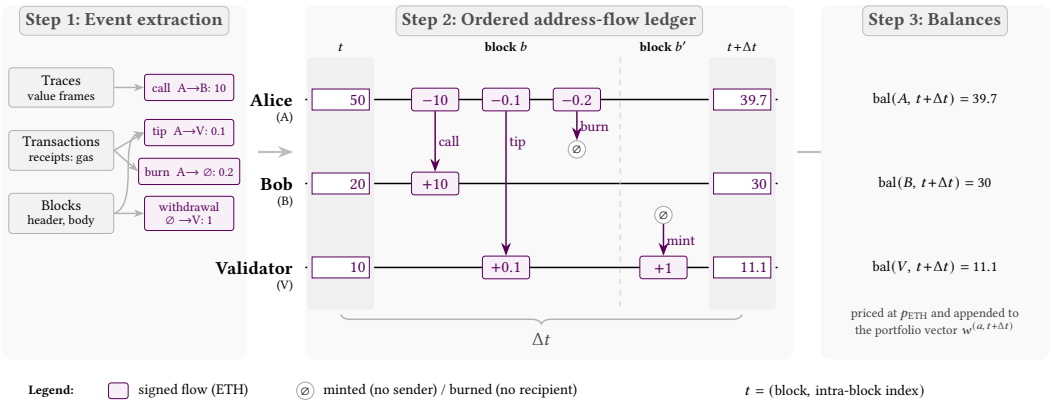


Fig. 29. Detailed native-ETH route of Figure 3 for three example accounts (Alice, Bob, and a validator V), with amounts in ETH. **Step 1** (left): balance-changing events are extracted from three source tables. Execution traces provide value-bearing call frames (here a call of 10 ETH from A to B). Transaction receipts, together with the block header, provide the gas fee, split into the priority tip paid to the block producer (0.1 ETH to V) and the burned base fee (0.2 ETH). Block bodies provide beacon-chain withdrawals (1 ETH to V). Call types that move no value and the subtrees of reverted calls are excluded, while fees are always kept. **Step 2** (centre): each event becomes signed entries on the lanes of its sender and recipient, ordered by block and intra-block index. Minted value ($\varnothing \to$) has no sender and burned value ($\to \varnothing$) no recipient, so the sum of all balances equals minted minus burned value. Adding the flows within Δt to the balances at t gives the balances at $t + \Delta t$, for Alice $50 - 10 - 0.1 - 0.2 = 39.7$. **Step 3** (right): the balances are priced with the daily ETH closing price in USD and enter the portfolio weight vector $w^{(a,t+\Delta t)}$ as an additional asset. In our analysis, Δt corresponds to one month.

Source tables. Three table families of the archive-node export enter the reconstruction: execution *traces* (one row per EVM execution frame with sender, recipient, value, frame type, and error status), *transactions* with their receipts (used only for fee computation), and *blocks* (headers with base fee and producer, bodies with beacon-chain withdrawals). Traces rather than the transaction list are the source of value movements because most ETH movement is internal: a call into a contract may trigger many nested calls, each of which can carry value, whereas the transaction table records only the outermost transfer. Since the root trace of every transaction is that external transfer, the transaction-level value field is never read: using both would double-count.

Event taxonomy. The central claim of flow-based reconstruction is completeness: native ETH changes hands in exactly the ways listed in Table 19, and in no others. Value-bearing `CALL`, `CREATE`, and `SELFDESTRUCT` frames cover ordinary transfers, payments into and between contracts, creation endowments, and balance sweeps. Three look-alike opcodes move nothing and are excluded: `DELEGATECALL` and `STATICCALL` carry no value, and `CALLCODE` may report one but executes as a transfer from the caller to itself. Every executed transaction debits its sender by gas consumed

times the effective gas price, but the fee's destination changed twice: the full fee went to the miner before the London fork; after EIP-1559 the base-fee share is burned and only the priority tip reaches the block producer; and since EIP-4844 blob-carrying transactions additionally pay a fully burned blob fee outside the effective gas price. Three event classes bypass execution entirely: the genesis allocation of 72,009,990.499 ETH, pre-Merge block and uncle rewards, and post-Shapella staking withdrawals from the beacon chain; the latter appear only in block bodies (in Gwei), so that execution data alone would silently lose all staking income from 2023 onward. The DAO fork, finally, moved roughly 12M ETH to a refund contract by an irregular state change outside any transaction. Reverted frames roll back their value transfers but not the fee, so we exclude the entire subtree of any errored frame while keeping its fee rows. Minted value (genesis, rewards, withdrawals) carries no sender and burned value (base and blob fees) no recipient, so at every block the sum of all ledger balances equals cumulative minted minus cumulative burned value.

ETH has no identity. Ethereum records only a balance per account, so units of ETH carry no identity, unlike the coins of Bitcoin's UTXO model, and no operation returns "the same" ETH, only an amount. A staking deposit is an ordinary transfer into the beacon deposit contract, which keeps every deposit, whereas a withdrawal credits newly minted ETH to the withdrawal address and draws nothing from that contract. In the ledger, a deposit and its later withdrawal are therefore two unrelated events, a transfer and a mint. The principal of withdrawn stake appears twice in the sum of all balances, once in the deposit contract and once at the withdrawal address, so the ledger's minted value exceeds ETH's issuance, and we do not count the deposit contract as a holder in our wealth and portfolio statistics (Section B.1.1).

Event class	Sender	Recipient	Amount source
Value call / create / self-destruct	trace	trace	trace value (wei)
Fee, priority tip	tx sender	block producer	computed from receipt
Fee, burned share (EIP-1559/4844)	tx sender	— (burn)	computed from receipt
Block / uncle reward (pre-Merge)	— (mint)	trace	trace value
Beacon withdrawal (post-Shapella)	— (mint)	block body	block body ($\times 10^9$)
Genesis allocation	— (mint)	genesis state	genesis state
DAO-fork state change	trace	trace	trace value

Table 19. Taxonomy of native-ETH balance-changing events and the physical origin of each ledger field. "—" denotes minting (no sender) or burning (no recipient). Beacon withdrawals are denominated in Gwei in the block body and converted to wei.

Intra-block ordering. Because an address may receive and re-spend funds inside one block, events are totally ordered within each block by a per-block sequence number: protocol events first, then per transaction its value transfers in depth-first trace order, followed by its fee burn and tip. The resulting transfer ledger covers blocks 0–23,918,198 (genesis through December 2025).

E.2 Execution Environment

All large-scale stages (ledger construction from raw logs and execution traces, portfolio reconstruction, and the downstream MPT optimisation/backtesting) were executed on an academic HPC cluster (details withheld for double-blind review), on CPU nodes with 128 cores and 512 GB RAM each.

Storage model. Large parquet datasets are stored on the cluster's parallel file system, which automatically promotes frequently accessed files from disk to a flash tier.

E.3 Computational Pipeline

The pipeline consists of four stages: building a data lake from raw Ethereum blockchain data, building per-asset ledgers for the Ethereum request for comments 20 (ERC-20) tokens and for native ETH, reconstructing per-account portfolio snapshots, and running MPT optimisation with forward-looking backtesting.

E.3.1 Data lake infrastructure. As a foundation to the reconstruction procedure, we utilise a data lake solution to store the raw Ethereum data. The *logs*, execution *traces*, *transactions* with their receipts, and *blocks* are the primary inputs to our data lake, organised in partitions of 10 000 Ethereum blocks in an efficiently accessible *Parquet* format. Parquet files use efficient compression and allow fast access through a scannable column-oriented storage format. This makes the format well-suited for parallel processing and provides a systematic infrastructure for processing the data.

E.3.2 Token ledgers. The most storage-intensive phase involves extracting and parsing the complete Ethereum log history from block #0 to block #23,943,069 (Jul-30-2015 to Dec-05-2025, approximately ten years of data). We scan the complete Ethereum log history and extract transfer events for tokens matching our allow-list, yielding 1,977,546,094 transfer events across 4,859 discovered tokens, stored as per-token ledger Parquet files totalling 42.4 GB. The transfer volume is heavily concentrated: the single largest token (wETH) accounts for 698 M transfers (35.3% of the total), while the median token has only 25,030 transfers. The allow-list already applies the ERC-20, price, volume, and supply checks of Section 3.2 (steps 1 to 4 in Figure 2). Validating the 4,859 ledgers against on-chain balances (step 5) leaves 4,562 tokens, which enter the reconstruction stage.

The MapReduce approach (described in Section 3.3) proceeds in two MapReduce passes. The first pass scans the raw log partitions, decodes Transfer, Deposit, and Withdrawal events, and writes per-token shard files. If a batch exceeds its timeout, the scheduler automatically splits it in half and retries both halves, avoiding manual intervention on slow partitions. The second pass merges shards into a single sorted ledger per token, storing each transfer as two rows (a negative entry for the sender and a positive entry for the recipient) so that account balances at any block height can be recovered by cumulative summation.

E.3.3 Native ETH ledger. The ETH route consumes the trace, transaction, and block tables of the archive-node export. A builder processes the chain in one-million-block chunks (sub-chunked when memory demands) and, per chunk, unions in a single DuckDB query the value-bearing trace frames (reverted subtrees excluded), the fee rows computed from transactions joined with block headers, and the beacon withdrawals exploded from block bodies, applies the deterministic intra-block ordering of Section E.1, and writes one Hive-partitioned Parquet file per chunk with completion markers so that interrupted runs resume. A second, fully out-of-core DuckDB pass explodes the resulting transfer ledger into the address-flow layout of the token ledgers, bucketing rows by address hash into 64 sorted shards so that every address lives in exactly one file; before completion, the pass checks that the ledger sums to cumulative minted minus cumulative burned value. Address activity is heavily skewed (the single busiest address accounts for roughly 4% of all rows), so oversized buckets are sub-split rather than balanced by hashing.

E.3.4 Portfolio snapshots. Portfolio reconstruction reads the token and ETH ledgers, converts Ethereum block heights to datetimes via an internal mapping, aggregates balances at each of the 72 monthly snapshots (blocks 9,193,266 to 23,914,921) via DuckDB, enriches holdings with decimals and USD prices, and writes the snapshot Parquet. All balances for a given asset–account pair are computed in a single SQL pass using conditional sums (`SUM(CASE WHEN block <= b THEN val END)`) per snapshot, rather than issuing 72 separate queries. The account universe is streamed in

contiguous, address-sorted batches to a fixed pool of persistent worker processes, each holding its own DuckDB connection, so that the per-file address-range statistics of the sorted ledgers prune the files a batch has to scan. Due to the dataset scale, reconstruction is split into 5 runs by snapshot period, whose outputs are consolidated into one dataset partitioned by snapshot block: 3,316.2 M account–snapshot observations in total (198.2 GB of compressed Parquet), growing from 10.0 M in January 2020 to 90.0 M in December 2025.

Proxy contract resolution. A token contract may be deployed as a proxy that delegates execution to an underlying implementation. We resolve such proxies before applying the ERC-20 selector checks of Section 3.2, covering EIP-1167 minimal proxies, EIP-1967 Transparent/UUPS proxies, beacon proxies, ZeppelinOS "unstructured storage" proxies, and proxies that expose an `implementation()` method.

E.3.5 MPT optimisation and backtesting. The optimisation stage takes the reconstructed holdings snapshots and, for each of the 72 monthly snapshots and every account with at least two assets (native ETH and wETH counted as one), solves three constrained mean–variance problems: maximise return at the account's current volatility (MaxRet), minimise variance at the account's current expected return (MinVar), and maximise the Sharpe ratio (MaxSR). Two naive benchmarks (equal-weight and market-cap-weight) are computed alongside, yielding 6 strategy variants per account. For each strategy, the pipeline evaluates realised 20-day forward returns and computes CAPM alpha and beta against a market benchmark.

The input comprises 832.8 M such account–snapshot observations (853.0 M before native ETH and wETH are merged). Per-snapshot account counts range from 3,964,445 to 21,552,463 (median: 11,141,805), reflecting the growth of the Ethereum account base over the 2020–2025 observation window. Due to dataset scale, optimisation jobs are split by snapshot ranges and executed on the 128-core nodes with a pool of worker processes per job. Price histories for all assets in each snapshot's universe are pre-loaded into shared memory and distributed to workers via process-level initialisation, avoiding redundant I/O across the millions of per-wallet optimisations. The final output comprises 817,525,786 rows (98.2% of the input) across 72 partitioned Parquet files totalling 359.7 GB; the remaining accounts cannot be optimised because fewer than two of their assets have sufficient price history at the snapshot, their priced value is zero, or their batch timed out.

E.4 Software Stack

This subsection describes the key libraries and tools underlying the pipeline described in Section E.3.

We implement the pipeline in Python 3.11 using PyArrow as the foundational columnar backend and Polars for efficient batch transformations. Parquet datasets are accessed via streaming interfaces to avoid materialising full tables in memory. For large-scale aggregation and lightweight analytical queries, we additionally employ DuckDB as an embedded analytical database (with explicitly bounded memory settings for stability). This stack provides a practical balance between scalability (HPC-ready I/O throughput), reproducibility (deterministic parquet-based artifacts), and modularity (separable reconstruction and optimisation stages).

For transparency and reproducibility, we provide the full pipeline implementation at https://anonymous.4open.science/r/modern-portfolio-theory-in-the-wild-7F81/.